

CURRICULUM VITAE

(updated on June 13, 2025)

Bryan E. Shepherd

Office Address:

Vanderbilt University School of Medicine
Department of Biostatistics
2525 West End, Suite 11000
Nashville, TN 37203

Office Phone Number:

615-343-3496

Education

Brigham Young University, Provo, UT

B.S. in Statistics, 1999

University Honors and Magna Cum Laude

Minors in Mathematics and Botany

University of Washington, Seattle, WA

M.S. in Biostatistics, 2001

Ph.D. in Biostatistics, 2005

Dissertation: “Causal Inference in HIV Vaccine Trials: Sensitivity analyses comparing outcomes only existing in a subset chosen post-randomization”

Academic Appointments

Assistant Professor, Department of Biostatistics, Vanderbilt University School of Medicine,
2005-2011

Associate Professor (with tenure), Department of Biostatistics, Vanderbilt University
School of Medicine, 2011-2017

Professor (with tenure), Department of Biostatistics, Vanderbilt University School of
Medicine, 2017-present

Vice Chair of Research, Department of Biostatistics, Vanderbilt University School of
Medicine, 2017-2019

Vice Chair of Faculty Affairs, Department of Biostatistics, Vanderbilt University School of Medicine, 2019-present

Affiliate Appointments

Visiting Scientist, Fundacion Huesped, Buenos Aires, Argentina, summer 2013

Visiting Scientist, Fred Hutchinson Cancer Research Center, Seattle, summer 2016

Visiting Scholar (Volunteer Appointment), UC San Diego, summer 2019

Vanderbilt Institute for Global Health, 2006-present

Vanderbilt Tuberculosis Center, 2013-present

Director of the Data Science Core of the Tennessee Center for AIDS Research (TN-CFAR), 2015-2024

Associate Director of the Data Science Core of the Tennessee Center for AIDS Research (TN-CFAR), 2024-present

Secondary Appointment, Vanderbilt Department of Biomedical Informatics, 2021-present

Adjunct Professor in the Department of Mathematical Sciences, Bayero University Kano, Nigeria, 2024-present

Other Employment

Computer Programmer, Department of Statistics, Brigham Young University, 1997

SAS Programmer, McBride Biostatistical Research, Morristown, NJ, summer 1998.

Teaching Assistant, Department of Statistics, Brigham Young University, 1998-1999

Research Assistant, Kenya Medical Research Institute and University of Washington, Nairobi, Kenya, summer 2000.

Teaching Assistant, Department of Biostatistics, University of Washington, 2000-2001

Research Assistant, Department of Biostatistics, University of Washington, 2002-2005

Professional Organizations

American Statistical Association

Chapter Representative, Middle Tennessee Chapter, 2006-2008

International Biometric Society

Professional Activities

Intramural

Chair of Allocation Committee, 2005-2012
Department of Biostatistics, Vanderbilt University

Member of Faculty Search Committee, 2005-present
Department of Biostatistics, Vanderbilt University

Member of Graduate Program Committee, 2008-present
Department of Biostatistics, Vanderbilt University

Member of Biostatistics Faculty Appointments and Promotions Committee, 2011-present
Department of Biostatistics, Vanderbilt University

Member of Biostatistics Promotions Committee, 2014-2018
Department of Biostatistics, Vanderbilt University

Member of Admissions Committee, 2010-2014
Department of Biostatistics, Vanderbilt University

Member of Vanderbilt Clinical and Translational Research Scholar program (VCTRS/VPST)
Selection Committee, 2012-present

Reviewer for Vanderbilt Edge for Scholars program, 2013-present

Chair of Publication Award Committee, 2014-18
Department of Biostatistics, Vanderbilt University

Member of the Vanderbilt Biostatistics Summer Internship Program for Undergraduates,
Department of Biostatistics, 2022-present

Member of the Vanderbilt School of Medicine Faculty Appointments and Promotions
Committee, 2023-present

Extramural

Ad hoc reviewer for The Bill and Melinda Gates Foundation, 2007

Member of Technical Expert Panel, 2010

“Future Research Priorities in Clinically Localized Prostate Cancer,” Blue Cross Blue Shield Association project sponsored by Agency for Healthcare Research and Quality

Member of Scientific Organizing Committee, “Causal Inference in Health Research,”
Centre de Recherche Mathematiques, Montreal, Canada, May 9-13, 2011

Ad hoc reviewer for Wellcome Trust, 2011

Ad hoc reviewer for National Institute of Allergy and Infectious Diseases, National Institutes of Health, “Targeted Clinical Research to Address Select Viral Infections,” April 2011

Ad hoc reviewer for National Institute of Allergy and Infectious Diseases, National Institutes of Health, “Targeting Inflammation and Immune Activation in HIV Disease,” October 2012

Ad hoc reviewer for the Medical Research Council, United Kingdom, November 2013

Chair of Data Safety and Monitoring Board, “Discontinuation of trimethoprim-sulfamethoxazole prophylaxis in adults on antiretroviral therapy in Kenya: a randomized trial,” 2012-2013

Member of Data Safety and Monitoring Board, “CAPRISA 008: Open-label randomized controlled trial to assess the implementation effectiveness and safety of 1% tenofovir gel provision through family planning services in KwaZulu-Natal, South Africa,” 2012-2015

Board of Directors, Nashville Health Management Foundation, 2014-2016

Member of Data Safety and Monitoring Board, “Providing Online Counseling for Home-Based HIV Testing with MSM Couples,” 2015-2020

Ad hoc reviewer for National Institutes of Health, “National Institute of Allergy and Infectious Diseases Special Emphasis Panel: NIAID Clinical Trial Planning Grant (R34) and Clinical Trial Implementation Cooperative Agreement (U01),” January 2015

Ad hoc reviewer for National Institutes of Health, “National Institute of Allergy and Infectious Diseases Special Emphasis Panel: Investigator Initiated Clinical Trial Applications (R34) and NIAID Clinical Trial Implementation Cooperative Agreement (U01),” December 2015

Ad hoc reviewer for Medical Research Council, January 2017

Chair of Data Safety and Monitoring Board, “Mobile WACHX: Evaluation of mHealth strategies to optimize adherence and efficacy of PMTCT/ART,” 2016-2019

Ad hoc reviewer for the National Institutes of Health, AIDS Clinical Epidemiology Study Section, November 2017

Ad hoc reviewer for the Medical Research Council, United Kingdom, January 2018

Ad hoc reviewer for Washington University at St. Louis Clinical and Translational Research Funding Program: Innovative analytical methodology projects, October 2018

Ad hoc reviewer for the National Institutes of Health, Special Emphasis Panel: Member conflict: AIDS and Related Research, March 2019

Ad hoc reviewer for the Medical Research Council, United Kingdom, April 2020

External Review Team, Department of Biostatistics, Yale School of Public Health, December 2020

Ad hoc reviewer for the National Institutes of Health, Population and Public Health Approaches to HIV/AIDS Study Section, March 2021

Member of Data Safety and Monitoring Board, “Immediate Fast-Track versus Standard Care for Persons Living with HIV in Haiti: a Randomized Pilot Study,” 2018-present

Member of the Safety Monitoring Committee for the Adolescent Medicine Trials Network (ATN) for HIV/AIDS Interventions, 2019-2022.

Ad hoc reviewer for the National Institutes of Health, Fogarty HIV Research Training Program for Low- and Middle-Income Country Institutions, December 2022

Ad hoc reviewer for the National Institutes of Health, Population and Public Health Approaches to HIV/AIDS, July 2023

Chair of Data Safety and Monitoring Board, “The HIV, Adipose Tissue Immunology, and Metabolism (HATIM) Study,” 2019-present.

Co-chair of Methods Working Group, North American AIDS Cohort Collaboration on Research and Design (NA-ACCORD), 2019-present.

Member of Data Safety Monitoring Board, HLB-SIMPLLe, 2023-present.

Associate Editor for the *American Journal of Epidemiology*, 2016-present

Associate Editor for *Biometrics*, 2018-present

Have reviewed papers for the following journals

AIDS

AIDS Research and Human Retroviruses

Australian and New Zealand Journal of Statistics

American Journal of Epidemiology

American Journal of Drug and Alcohol Abuse

American Journal of Public Health

Annals of Applied Statistics

Biometrical Journal

Biometrics

Biometrika

Biostatistics

Blood

BMC Health Services Research

BMC Infectious Diseases

BMC Medical Research Methodology

Canadian Journal of Statistics

Communication in Statistics - Simulation and Computation

Contemporary Clinical Trials

Clinical Trials

Drug and Alcohol Dependence

Epidemiologic Reviews

Epidemiology

Methods of Information in Medicine

International Journal of Biostatistics

International Journal of Epidemiology

International Journal of Women's Health

International Statistical Review

Journal of Acquired Immune Deficiency Syndromes

Journal of Applied Statistics

Journal of Biopharmaceutical Statistics

Journal of Clinical and Translational Science

Journal of Clinical Epidemiology

Journal of Computational and Graphical Statistics

Journal of Infectious Diseases

Journal of Statistical Software

Journal of the American Medical Association

JAMA Oncology

Journal of the American Statistical Association

Journal of the International AIDS Society

Journal of the Royal Statistical Society, Series B

Journal of the Royal Statistical Society, Series C

Lancet HIV
Nature Communications
Pharmacoepidemiology and Drug Safety (“Best Reviewer” 2018)
PLoS Computational Biology
PLoS ONE
PLoS Medicine
Proceedings of the Royal Society B: Biological Sciences
Scandinavian Journal of Statistics
Statistical Communications in Infectious Diseases
Statistics in Medicine
Statistical Methods in Medical Research
The American Statistician
Trials

Academic Awards

Trustees Scholarship, 1993, Brigham Young University
 Edwin Hinckley Scholarship, 1998, Brigham Young University
 Undergraduate Research Award, 1998, Brigham Young University
 Graduate Merit Award, 1999, Department of Biostatistics, University of Washington
 Senior Student Award, 2005, Department of Biostatistics, University of Washington
 Shayle Searle Fellow, 2016, School of Mathematics and Statistics, Victoria University
 of Wellington, New Zealand

 Outstanding Faculty Mentor, 2016, Vanderbilt Department of Biostatistics
 Canadian Journal of Statistics Award, 2016, Statistical Society of Canada
 Methods Publication Award, 2016, Vanderbilt Department of Biostatistics
 Methods Publication Award, 2017, Vanderbilt Department of Biostatistics
 Fellow of the American Statistical Association, 2018
 Methods Publication Award, 2021, Vanderbilt Department of Biostatistics
 MERIT (Method to Extend Research in Time) Award, 2023, U.S. National Institutes
 of Health
 Best Paper in *Biometrics* by an IBS Member, 2023, International Biometric Society
 Methods Publication Award, 2023, Vanderbilt Department of Biostatistics

Teaching Activities

Courses

Introduction to Probability and Statistical Theory

- Fall 2011
- Fall 2012
- Fall 2013
- Fall 2014
- Fall 2015
- Fall 2016
- Fall 2017

Causal Inference

- Spring 2018
- Spring 2020
- Spring 2022
- Fall 2024

Individualized Courses for Specific Students

Advanced Methods for Epidemiology (Epidemiology 332) (Spring 2012); Lan Zhang; *Respondent Driven Sampling*

Independent Study in Biostatistics (Biostatistics 393) (Fall 2011 - Spring 2012); Nate Mercaldo; *Dynamic optimal treatment regimens applied to initiating therapy in HIV*

Advanced Methods for Epidemiology (Epidemiology 332) (Spring 2021); Cassandra Oliver; *Analysis of Cluster Data*

Independent Study in Biostatistics (Spring 2025); Alexis Fleming; *Advanced methods for causal inference and machine learning*

Research Supervision

Primary Advisor:

Valentine Wanga (Biostatistics MS 2014, took position at Institute for Health Metrics and Evaluation, Seattle, WA)

Qi Liu (Biostatistics PhD 2016, took position at Merck, Rahway, NJ)

Lingjun Fu (Biostatistics MS 2017, took position at Amazon, Seattle, WA)

Mark Giganti (Biostatistics PhD 2018, took position at Center for Biostatistics in AIDS Research, Harvard, Boston, MA)

Svetlana Eden (Biostatistics PhD, 2020, took position at Vanderbilt University Medical Center, Nashville, TN)

Sarah Lotspeich (Biostatistics PhD, 2021, took position at University of North Carolina Chapel Hill)

Yuqi Tian (Biostatistics PhD, 2022, took position at Uber, San Francisco, CA)

Shengxin Tu (Biostatistics PhD, 2024, took position at US Food and Drug Administration, Washington DC)

Julia Thome (Biostatistics PhD, 2024, took position at Johns Hopkins University, Baltimore MD)

Joshua Slone (Biostatistics PhD student, 2022-present)

Jeffrey Liang (Biostatistics PhD student, 2023-present)

PhD Student Committees:

Sangsoo Park (Economics PhD 2008)

Jisong Wu (Economics PhD 2009)

Stephany Duda (Biomedical Informatics PhD 2011)

Heng Chen (Economics PhD 2012)

Lan Zhang (Epidemiology PhD 2013)

Webster Kasongo (Epidemiology PhD 2013)

Xi Huo (Mathematics PhD 2014)

Jun Tao (Epidemiology PhD 2016)

Laurie Samuels (Biostatistics PhD 2016)

Chen Zhang (Epidemiology PhD 2017)

Nate Mercaldo (Biostatistics PhD 2017)

Harold Chiang (Economics PhD 2020)

Ryan Jarrett (Biostatistics PhD 2021)

Daniel Sacks (Epidemiology PhD 2022)

Cassandra Oliver (Epidemiology PhD 2022)

Nathan James (Biostatistics PhD 2022)

Michael Williams (Biostatistics PhD 2022)

Jacqueline O'Neal (Biostatistics PhD 2022)

Elizabeth Sigworth (Biostatistics PhD 2022)

Chiara Di Gravio (Biostatistics PhD 2023)

Jamie Joseph (Biostatistics PhD 2023)

Faculty Mentoring Committees:

Pingsheng Wu (Research Assistant Professor, Department of Medicine)
Carolyn Audet (Assistant/Associate Professor, Institute for Global Health)
April Pettit (Assistant Professor, Division of Infectious Diseases)
Celestine Wanjalla (Research Fellow, Division of Infectious Diseases)
Peter Rebeiro (Assistant Professor, Division of Infectious Diseases)
James Carlucci (Assistant Professor, Pediatric Infectious Diseases)
Ran Tao (Assistant/Associate Professor, Biostatistics)
Andrew Spieker (Assistant Professor, Biostatistics)
Gustavo Amorim (Assistant Professor, Biostatistics)
Svetlana Eden (Research Assistant Professor, Biostatistics)
Qingyan Xiang (Assistant Professor, Biostatistics)

International Trainees:

Yanink Caro-Vega (Fogarty International Mentoring Program, Mexico, 2017-2021)
Lara Coelho (Fogarty International Mentoring Program, Brazil, 2017-2021)
Ibrahim Yusuf Ibrahim (Vanderbilt Nigeria Biostatistics Training Program, 2022-2023)
Isaac Fwemba (Vanderbilt-Emory-Cornell-Duke Training Program, 2022-2023)
Abdurrahman Abdulhamid (Vanderbilt Nigeria Biostatistics Training Program, 2022-2023)
Safiya Sada (Vanderbilt Nigeria Biostatistics Training Program, 2023-2024)
Rukayya Alkassim (Vanderbilt Nigeria Biostatistics Training Program, 2023-2024)
Habebah Kakudi (Vanderbilt Nigeria Biostatistics Training Program, 2024-2025)

Undergraduate Trainees:

Joseph Sherrill (summer 2019)
Terrence Smith (Vanderbilt Biostatistics Summer Internship Program for Undergraduates, 2023)
Naij Taylor (Vanderbilt Biostatistics Summer Internship Program for Undergraduates, 2024)

Other Mentoring:

James Carlucci (Medical School Emphasis project, 2007)
Tera Howard (Medical School Emphasis project, 2008)
Diana Lemly (Medical School Emphasis project, 2008)
Keipp Talbot (Infectious Diseases Fellow, 2008-2009)
Vlada Melekhin (Infectious Diseases Fellow, 2007-2011)
Yuri Van der Heijden (Infectious Diseases Fellow, 2012-2016)

Internal Seminars, Internal Lectures, and Workshops

1. Stochastic modeling of hematopoiesis.
Department of Biostatistics, Vanderbilt University, November 9, 2005
2. An analysis problem arising in HIV vaccine trials and our proposed solution.
Infectious Disease Grand Rounds, Vanderbilt University, February 1, 2006
3. The causal effect of finasteride on the severity of prostate cancer.
Department of Biostatistics, Vanderbilt University, September 20, 2006
4. Baseline CD4 lymphocyte percentage and HIV/AIDS disease progression on therapy: the promise and problems of predictive modeling.
Vanderbilt-Meharry Center for AIDS Research Work in Progress Seminar, May 24, 2007
5. Guest lectures to Vanderbilt Master of Science in Clinical Investigation Program
Design of medical studies, September 20, 2007
Some analysis advice, September 24, 2008
6. Sensitivity analyses comparing time-to-event outcomes only existing in a subset selected post-randomization, with application to HIV vaccine trials.
Department of Economics, Vanderbilt University, Statistics Study Group Seminar Series, February 14, 2008
7. Biostatistics 101.
2nd Annual Administrative Retreat, Department of Biostatistics, Vanderbilt University, April 4, 2008
8. The cost of checking proportional hazards.
Department of Biostatistics, Vanderbilt University, April 23, 2008
9. An introduction to linear mixed effects models.
MS Continuing Education Meeting, Department of Biostatistics, Vanderbilt University, April 17 and May 1, 2008

10. Statistical methods to account for data errors discovered from an audit.
Department of Biostatistics, Vanderbilt University, July 15, 2009
11. This American Statistician's Life.
School for Math and Science at Vanderbilt, December 3, 2009
12. Test of Association between Two Ordinal Variables while Adjusting for Covariates.
Department of Mathematics, Vanderbilt University, Biomathematics Group, April 6, 2010
13. Multiple imputation strategies for handling missing data.
National Institute of Mental Health International HIV/AIDS Conference, Nashville, June 17, 2010
14. Presentations to the James Crowe Jr. Lab, Vanderbilt University
Power and sample size, September 8, 2006
T-tests and Wilcoxon rank sum tests, February 2, 2007
Power and sample size, June 21, 2007
Linear regression, December 7, 2007
Analyzing neut data, June 6, 2008
Analyzing correlated data, December 5, 2008
Sample size and power calculations, May 15, 2009
Assessing the effect of vaccination on post-infection outcomes, October 30, 2009
Statistical issues to consider in translational research, March 19, 2010
15. Ordered logistic regression.
Guest Lecture to Epidemiology Graduate Students: November 9, 2010; October 27, 2011; October 25, 2012; October 17, 2013; November 20, 2014; November 1, 2016; October 24, 2017; October 30, 2018; October 22, 2019; October 7, 2021; September 28, 2022; October 31, 2023; September 24, 2024.
16. Estimating the optimal CD4 count for HIV-infected individuals to start antiretroviral therapy.
Department of Biostatistics, Vanderbilt University, January 12, 2011
17. Estimating the optimal CD4 count for HIV-infected individuals to start antiretroviral therapy.
Vanderbilt Institute for Global Health Faculty Meeting, January 25, 2011
18. Design of Medical Studies and Some Analysis Advice.
Vanderbilt Institute for Global Health Institute in Research Development and Ethics: May 16, 2011; August 17, 2012; October 18, 2013; September 19, 2014; October 29, 2015; October 13, 2016; October 6, 2017; October 9, 2018; October 9, 2019; October 5, 2022; October 6, 2023; October 15, 2024.
19. IeDEA Network Coordinating Center (INCC) at Vanderbilt. With Bill Wester and Mary Louise Lindegren. Vanderbilt Institute for Global Health Faculty Research Development Meeting, November 29, 2011

20. The Caribbean, Central and South America network for HIV Epidemiology: experiences from an observational HIV cohort. With Catherine McGowan.
Infectious Diseases Grand Rounds, Vanderbilt University, December 7, 2011
21. Contemporary statistical inference (Biostatistics 342); guest lecture.
January 18 and 20, 2012; January 16 and 18, 2013
22. Advanced statistical inference (Biostatistics 362); guest lecture.
February 13, 2013
23. Principal stratification: an analysis approach for dealing with post-randomization selection and truncation by death.
Health Services Research Work in Progress Seminar, Vanderbilt University, March 12, 2013
24. Probability-scale residuals.
Department of Biostatistics, Vanderbilt University, January 22, 2014
25. Comparing different methods for assessing when to start antiretroviral therapy.
Infectious Diseases Grand Rounds, Vanderbilt University, March 13, 2014
26. The Caribbean, Central and South American Network for HIV Research (CCASAnet).
Vanderbilt Institute for Global Health Faculty Research Development Meeting, May 5, 2014.
27. Incorporating partially validated data into analyses of error-prone datasets to improve estimation.
Center for Quantitative Sciences Seminar, April 17, 2015.
28. Methods for Research in Implementation Science.
Short course co-taught with Sten Vermund and Meridith Blevins as part of Vanderbilt-UEM Partnership for Research in Implementation Science Mozambique (PRISM).
Universidade Eduardo Mondlane. Maputo, Mozambique. December 5-9, 2016.
29. Modeling continuous response variables using ordinal regression.
Department of Biostatistics, Vanderbilt University, January 17, 2018.
30. TuLHIP Study: Tuberculosis Treatment Length and Adherence in HIV-positive Persons.
Centro de Investigacao em Saude de Manica, Mozambique. October 18, 2018.
31. Modeling continuous response variables using ordinal regression.
Continuing Education - Method Implementation Workshop. Department of Biostatistics, Vanderbilt University, August 16, 2019.
32. New statistical methods motivated and illustrated using data from Tennessee Center for AIDS Research (TN-CFAR) investigators.
Work in Progress Seminar. TN-CFAR. June 8, 2020.

33. Multi-wave validation sampling for error-prone electronic health records.
Department of Biostatistics, Vanderbilt University, October 5, 2022.
34. Regression.
Fogarty IeDEA Mentorship Program, virtual. November 17, 2022.
35. Linear, logistic, and ordinal regression models with R statistical software.
Vanderbilt-Nigeria Biostatistics Workshop. Aminu Kano Teaching Hospital, Kano, Nigeria. June 12-16, 2023. (with Chun Li)
36. What is a biostatistician? Brigham Young University Statistics Students Club, virtual, September 19, 2023.
37. Survival analysis with R statistical software.
Vanderbilt-Nigeria Biostatistics Workshop. Aminu Kano Teaching Hospital, Kano, Nigeria. June 3-7, 2024. (with Ben French)
38. Analysis with missing data.
Vanderbilt-Nigeria Biostatistics Workshop. Aminu Kano Teaching Hospital, Kano, Nigeria. June 2-6, 2025. (with Cindy Chen)

Research Program

R03 A1068069 PI: Talbot 02/01/07-01/31/09
Epidemiology and Clinical Features of Human Coronavirus
Role: Biostatistician

R01 DA023879 PI: Frangakis 07/15/07-06/30/11
Statistical Designs and Methods for Partially Controlled HIV/AIDS Studies
Role: Co-Investigator / PI of Vanderbilt subcontract

R01 DA026207 PI: Markham 09/01/08-07/31/10
Effect of cocaine and an LTR polymorphism on HIV-1 pathogenesis
Role: Co-investigator

APW200653698/SR-1400313190 PI: Vermund 08/01/12-04/30/13
Factors Influencing Changes in Child Survival in sub-Saharan Africa from 1990 to 2009
Role: Co-Investigator

P30 AI054999 PI: Kalams 05/15/03-08/31/13
Vanderbilt Meharry Center For AIDS Research
Role: Co-Investigator

U01 AI069918	PI: Moore	07/01/13-06/30/15
NA-ACCORD Supplement; obesity and ACE-inhibitors (Vanderbilt PI: Sterling)		
Role: Co-Investigator		
VUMC 41254	PI: Sterling	01/01/13-06/30/14
Bringing K-RITH into IeDEA		
Role: Co-Investigator		
R01 AI094562	PI: Vermund	03/15/11-02/28/15
Multi-component HIV Intervention Packages for Chinese MSM		
Role: Co-Investigator		
R34 AI091446	PI: Vermund	08/01/11-07/31/14
Expanding Testing, Linkage, and Treatment for AIDS Prevention Among MSM in China		
Role: Co-Investigator		
R01 HD075075	PI: Aliyu	09/28/12-06/30/14
Optimizing integrated PMTCT services in rural North-Central Nigeria		
Role: Co-Investigator		
U01 AI069923	PI: McGowan	06/15/06-present
Caribbean, Central and South America Network for HIV epidemiology (CCASAnet)		
Role: Co-Investigator		
P30 AI110527	PI: Mallal	04/01/15-present
Tennessee Center for AIDS Research (TN-CFAR)		
Role: Co-Investigator and Director of Biostatistics and Biomedical Informatics Core		
K08 AI106420	PI: Van der Heijden	08/01/14-07/31/19
Fluoroquinolone Resistance in Patients with Multidrug-Resistant Tuberculosis		
Role: Co-Investigator		
R24 AI124872	PI: Duda	04/30/16-03/31/21
Harmonist: A Scalable Toolkit for Standardizing and Coordinating Data Sharing Across International Research Networks		
Role: Co-Investigator		
R01 MH113478-01	PI: Audet	05/12/17-present
Partners-based HIV Treatment for Sero-concordant Couples attending Antenatal Care		
Role: Co-Investigator		

2. Mwachari CW, Shepherd BE, Cleopa O, Odhiambo JA, Cohen CR. Mortality and burden of disease in a cohort of HIV-seropositive adults in Nairobi, Kenya. *International Journal of STD and AIDS* 2004; 15: 120–126.
3. Shepherd BE, Guttorp P, Lansdorp PM, Abkowitz JL. Estimating human hematopoietic stem cell kinetics using granulocyte telomere lengths. *Experimental Hematology* 2004; 32: 1040–1050.
4. Gilbert PB, Ackers M, Berman PW, Francis DP, Popovic V, Hu D, Heyward WL, Sinangil F, Shepherd BE, Gurwith M. HIV-1 virologic and immunologic progression and antiretroviral therapy initiation among HIV-1-infected participants in an efficacy trial of recombinant glycoprotein 120 vaccine. *Journal of Infectious Diseases* 2005; 192: 974–983.
5. Shepherd BE, Rossini AJ, Soto RJ, de Rivera IL, Mullins JI. Sampling designs for HIV molecular epidemiology with application to Honduras. *AIDS Research and Human Retroviruses* 2005; 21: 907–914.
6. Shepherd BE, Gilbert PB, Jemai Y, Rotnitzky A. Sensitivity analyses comparing outcomes only existing in a subset selected post-randomization, conditional on covariates, with application to HIV vaccine trials. *Biometrics* 2006; 62: 332–342.
7. Parham GP, Sahasrabudhe VV, Mwanahamuntu MH, Shepherd BE, Hicks ML, Stringer EM, Vermund SH. Prevalence and predictors of squamous intraepithelial lesions of the cervix in HIV-infected women in Lusaka, Zambia. *Gynecologic Oncology* 2006; 103: 1017–1022.
8. Drake WP, Dhason MS, Nadaf M, Shepherd BE, Vivedulu S, Hajizadeh R, Newman LS, Kalams SA. Cellular recognition of Mycobacterium ESAT-6 and katG peptides in systemic sarcoidosis. *Infection and Immunity* 2007; 75: 527–530.
9. Hulgán T, Shepherd BE, Raffanti SP, Fusco JS, Beckerman R, Barkanic G, Sterling TR. Absolute and percentage CD4+ lymphocytes are independent predictors of disease progression in HIV-infected persons initiating HAART. *Journal of Infectious Diseases* 2007; 195: 425–431.
10. Tian C, Luskin GK, Dischert KM, Higginbotham JN, Shepherd BE, Crowe JE Jr. Evidence for preferential Ig gene usage and differential TdT and exonuclease activities in human naive and memory B cells. *Molecular Immunology* 2007; 44: 2173–83.
11. Shepherd BE, Gilbert PB, Mehrotra DV. Eliciting a counterfactual sensitivity parameter. *The American Statistician* 2007; 61: 56–63.
12. Shepherd BE, Gilbert PB, Lumley T. Sensitivity analyses comparing time-to-event outcomes only existing in a subset selected post-randomization. *Journal of the American Statistical Association* 2007; 102: 573–582.

13. Hajizadeh R, Sato H, Carlisle J, Nadaf MT, Evans W, Shepherd BE, Miller RF, Kalams SA, Drake WP. Mycobacterium tuberculosis antigen 85A induces Th-1 immune responses in systemic sarcoidosis. *Journal of Clinical Immunology* 2007; 27: 445–454.
14. Williams JV, Chen Z, Cseke G, Wright DW, Keefer CJ, Tollefson SJ, Hessell A, Podsiad A, Shepherd BE, Sanna PP, Burton DR, Crowe JE Jr, Williamson RA. A Recombinant Human Monoclonal Antibody to Human Metapneumovirus Fusion Protein that Neutralizes Virus in Vitro and is Effective Therapeutically in Vivo. *Journal of Virology* 2007; 81: 8315–8324.
15. Shepherd BE, Kiem HP, Lansdorp PM, Aubert G, Dunbar CE, LaRochelle A, Seggewiss R, Guttorp P, Abkowitz JL. Hematopoietic stem cell behavior in non-human primates. *Blood* 2007; 110: 1806–1813.
16. Mok H, Lee S, Utley TJ, Shepherd BE, Polosukhin VV, Collier ML, Davis NL, Johnston RE, Crowe JE Jr. Venezuelan equine encephalitis replicon particles encoding respiratory syncytial virus surface glycoproteins induce protective mucosal responses in mice and cotton rats. *Journal of Virology* 2007; 81: 13710–13722.
17. Jemai Y, Rotnitzky A, Shepherd BE, Gilbert PB. Semiparametric estimation of treatment effects given baseline covariates on an outcome measured after a post-randomization event occurs. *Journal of the Royal Statistical Society, Series B* 2007; 69: 879–901.
18. Riviello ED, Sterling TR, Shepherd B, Fantan T, Makhema J. HIV in the workplace in Botswana: incidence, prevalence, and disease severity. *AIDS Research and Human Retroviruses* 2007; 23: 1453–1460.
19. Carlisle J, Evans W, Hajizadeh R, Shepherd BE, Ott RD, Richter K, Drake WP. Multiple Mycobacterium antigens induce interferon-gamma production from sarcoidosis peripheral blood mononuclear cells. *Journal of Clinical and Experimental Immunology* 2007; 150: 460–468.
20. Tian C, Luskin GK, Dischert KM, Higginbotham JN, Shepherd BE, Crowe JE Jr. Immunodominance of the VH1-46 antibody gene segment in the primary repertoire of human rotavirus-specific B cells is reduced in the memory compartment through somatic mutation of nondominant clones. *Journal of Immunology* 2008; 180: 3279–3288.
21. Shepherd BE. The cost of checking proportional hazards. *Statistics in Medicine* 2008; 27: 1248–1260.
22. Carlucci JG, Kamanga A, Sheneberger R, Shepherd BE, Jenkins CA, Spurrier J, Vermund SH. Predictors of adherence to antiretroviral therapy in rural Zambia. *Journal of Acquired Immune Deficiency Syndromes* 2008; 47: 615–622.

23. Allen SS, Evans W, Carlisle J, Hajizadeh R, Nadaf M, Pride DT, Shepherd BE, Johnson JE, Drake WP. Superoxide dismutase A antigens derived from molecular analysis of sarcoidosis granulomas elicit systemic Th-1 immune responses. *Respiratory Research* 2008; 9: 36.
24. Utley TJ, Ducharme NA, Varthakavi V, Shepherd BE, Santangelo PJ, Lindquist ME, Goldenring JR, Crowe JE Jr. Respiratory syncytial virus uses a Vps4-independent budding mechanism controlled by Rab11-FIP2. *Proceedings of the National Academy of Sciences* 2008; 105: 10209–10214.
25. Morel KR, Shepherd BE. Developing a symptom validity test for posttraumatic stress disorder: application of the binomial distribution. *Journal of Anxiety Disorders* 2008; 22: 1297–1302.
26. Mok H, Tollefson SJ, Podsiad AB, Shepherd BE, Polosukhin VV, Johnston RE, Williams JV, Crowe JE Jr. An alphavirus replicon-based human metapneumovirus vaccine is immunogenic and protective in mice and cotton rats. *Journal of Virology* 2008; 82: 11410–11418.
27. Shepherd BE, Redman MW, Ankerst DP. Does finasteride affect the severity of prostate cancer? A causal sensitivity analysis. *Journal of the American Statistical Association* 2008; 103: 1392–1404.
28. Tsai CS, Shepherd BE, Vermund SH. Does douching increase risk for sexually transmitted infections? A prospective study in high-risk adolescents. *American Journal of Obstetrics and Gynecology* 2009; 200: 38.e1–8.
29. Pettit AC, Barkanic G, Stinnette S, Rebeiro P, Blackwell R, Raffanti SP, Shepherd BE, Sterling TR. Potentially-preventable tuberculosis among HIV-infected persons in the era of highly active antiretroviral treatment. *International Journal of Tuberculosis and Lung Disease* 2009; 13: 355–359.
30. Aliferis CF, Statnikov A, Tsamardinos I, Schildcrout JS, Shepherd BE, Harrell FE Jr. Factors influencing the statistical power of complex data analysis protocols for molecular signature development from microarray data. *PLoS ONE* 2009; 4: e4922.
31. Lemly DC, Shepherd BE, Hulgan T, Rebeiro P, Stinnette S, Blackwell RB, Bebawy S, Kheshti A, Sterling TR, Raffanti SP. Race and sex differences in antiretroviral therapy use and mortality among HIV-infected persons in care. *Journal of Infectious Diseases* 2009; 199: 991–998.
32. Shepherd BE, Sterling TR, Moore RD, Raffanti SP, Hulgan T. Cross-cohort heterogeneity encountered while validating a model for HIV disease progression among antiretroviral initiators. *Journal of Clinical Epidemiology* 2009; 62: 729–737.
33. Tuboi SH, Schechter M, McGowan CC, Cesar C, Krolewiecki A, Cahn P, Wolff M, Pape JW, Padgett D, Madero JS, Gotuzzo E, Masys DR, Shepherd BE. Mortality during the first year of potent antiretroviral therapy in HIV-1-infected patients in

- 7 sites throughout Latin America and the Caribbean. *Journal of Acquired Immune Deficiency Syndromes* 2009; 51: 615–623.
34. Talbot HK, Shepherd BE, Crowe JE Jr, Griffin MR, Edwards KM, Podsiad AB, Tollefson SJ, Wright PF, Williams JV. The pediatric burden of human coronaviruses evaluated for twenty years. *The Pediatric Infectious Disease Journal* 2009; 28: 682–687.
 35. Oswald-Richter KA, Culver DA, Hawkins C, Hajizadeh R, Abraham S, Shepherd BE, Jenkins CA, Judson MA, Drake WP. Cellular responses to mycobacterial antigens are present in sarcoidosis diagnostic bronchoalveolar lavage. *Infection and Immunity* 2009; 77: 3740–3748.
 36. Li AH, Phanuphak N, Sahasrabuddhe VV, Chaithongwongwatthana S, Vermund SH, Jenkins CA, Shepherd BE, Teeratakulpisarn N, van der Lugt J, Avihingsanon A, Ruxrungtham K, Shikuma C, Phanuphak P, Ananworanich J. Anal squamous intraepithelial lesions among HIV-positive and HIV-negative men who have sex with men in Thailand. *Sexually Transmitted Infections* 2009; 85: 503–507.
 37. Melekhin VV, Shepherd BE, Stinnette SE, Rebeiro PF, Barkanic G, Raffanti SP, Sterling TR. Antiretroviral therapy initiation before, during, or after pregnancy in HIV-1-infected women: maternal virologic, immunologic, and clinical response. *PLoS ONE* 2009; 4: e6961.
 38. Sahasrabuddhe VV, Bhosale RA, Joshi SN, Kavatkar AN, Nagwanshi CA, Kelkar RS, Jenkins CA, Shepherd BE, Sahay S, Risbud AR, Vermund SH, Mehendale SM. Prevalence and predictors of colposcopic-histopathologically confirmed cervical intraepithelial neoplasia in HIV-infected women in India. *PLoS ONE* 2010; 5: e8634.
 39. Oswald-Richter K, Sato H, Hajizadeh R, Shepherd BE, Sidney J, Sette A, Newman LS, Drake WP. Mycobacterial ESAT-6 and katG are Recognized by Sarcoidosis CD4+ T Cells When Presented by the American Sarcoidosis Susceptibility Allele, DRB1*1101. *Journal of Clinical Immunology* 2010; 30:157–166.
 40. Heimburger DC, Koethe JR, Nyirenda C, Bosire C, Chiasaera JM, Blevins M, Munoz AJ, Shepherd BE, Potter D, Zulu I, Chisembele-Taylor A, Chi BH, Stringer JSA, Kabagambe EK. Serum Phosphate Predicts Early Mortality in Adults Starting Antiretroviral Therapy in Lusaka, Zambia: A Prospective Cohort Study. *PLoS ONE* 2010; 5: e10687.
 41. Cesar C, Shepherd BE, Krolewiecki AJ, Fink VI, Schechter M, Tuboi SH, Wolff M, Pape JW, Leger P, Padgett D, Madero JS, Gotzso E, McGowan CC, Masys DR, Cahn PE. Rates and reasons for early change of first HAART in HIV-1-infected patients in 7 sites throughout the Caribbean and Latin America. *PLoS ONE* 2010; 5: e10490.
 42. Li C, Shepherd BE. Test of association between two ordinal variables while adjusting for covariates. *Journal of the American Statistical Association* 2010; 105: 612–620.

43. Shepherd BE, Jenkins CA, Rebeiro PF, Stinnette SE, Bebawy SS, McGowan CC, Hulgán T, Sterling TR. Estimating the optimal CD4 count for HIV-infected patients to start antiretroviral therapy. *Epidemiology* 2010; 21:698–705.
44. Melekhin VV, Shepherd BE, Jenkins CA, Stinnette SE, Rebeiro PF, Bebawy SS, Rasbach DA, Hulgán T, Sterling TR. Postpartum discontinuation of antiretroviral therapy and risk of maternal AIDS-defining events, non-AIDS-defining events, and mortality among a cohort of HIV-1-infected women in the United States. *AIDS Patient Care STDS* 2010; 24: 279–286.
45. Kazi AM, Shah SA, Jenkins CA, Shepherd BE, Vermund SH. Risk Factors and prevalence of tuberculosis, human immunodeficiency virus, syphilis, hepatitis B virus, and hepatitis C virus among prisoners in Pakistan. *International Journal of Infectious Diseases* 2010; 14S: e60–e66.
46. Wolff MJ, Cortes CM, Shepherd BE, Beltran CB. Long term outcomes of a national expanded access program to antiretroviral therapy: the Chilean AIDS Cohort. *Journal of Acquired Immune Deficiency Syndromes* 2010; 55: 368–374.
47. Thornburg NJ, Shepherd B, Crowe JE Jr. TGF- β is a major regulator of altered neonatal human immune responses following respiratory syncytial virus infection. *Journal of Virology* 2010; 84: 12895–12902.
48. Koethe JR, Blevins M, Nyirenda C, Kabagambe EK, Shepherd BE, Wester CW, Zulu I, Chiasera JM, Mulenga LB, Mwango A, Heimbürger DC. Nutrition and inflammation serum biomarkers are associated with 12-week mortality among malnourished adults initiating antiretroviral therapy in Zambia. *Journal of the International AIDS Society* 2011; 14: 19.
49. van der Heijden YF, Miller G, Wright PW, Shepherd BE, Daniels TL, Talbot TR. Clinical Impact of Blood Cultures Contaminated with Coagulase-negative Staphylococci at an Academic Medical Center. *Infection Control and Hospital Epidemiology* 2011; 32: 623–625.
50. Crabtree-Ramirez B, Caro-Vega Y, Shepherd BE, Wehbe F, Cesar C, Cortes C, Padgett D, Koenig S, Gotuzzo E, Cahn P, McGowan C, Masys D, Sierra-Madero J. Cross-sectional analysis of late HAART initiation in Latin America and the Caribbean: late testers and late presenters. *PLoS ONE* 2011; 6: e20272.
51. Wester CW, Koethe JR, Shepherd BE, Stinnette SE, Rebeiro PF, Kipp AM, Hong H, Bussmann H, Gaolathe T, McGowan CC, Sterling TR, Marlink RG. Non-AIDS defining events among HIV-1 infected adults receiving combination antiretroviral therapy in urban settings in Sub-Saharan Africa and the United States. *AIDS* 2011; 25: 1471–1479.
52. Qian HZ, Stinnette SE, Rebeiro PF, Kipp A, Shepherd BE, Samenow CP, Jenkins CA, No P, McGowan CC, Hulgán T, Sterling TR. The relationship between injection and non-injection drug use and HIV disease progression. *Journal of Substance Abuse Treatment* 2011; 41: 14–20.

53. Fink VI, Shepherd BE, Cesar C, Krolewiecki A, Wehbe F, Cortes CP, Crabtree-Ramirez B, Padgett D, Shafae M, Schechter M, Gotuzzo E, Bacon M, McGowan C, Cahn P, Masys D. Cancer in HIV-infected persons from the Caribbean, Central and South America. *Journal of Acquired Immune Deficiency Syndromes* 2011; 56: 467–473.
54. Pettit AC, Jenkins CA, Stinnette S, Rebeiro P, Blackwell R, Raffanti SP, Shepherd BE, Sterling TR. Tuberculosis risk before and after highly active antiretroviral therapy initiation: Does HAART increase the short-term TB risk in a low incidence TB setting? *Journal of Acquired Immune Deficiency Syndromes* 2011; 57: 305–310.
55. Shepherd BE, Gilbert PB, Dupont CT. Sensitivity analyses comparing time-to-event outcomes only existing in a subset selected postrandomization and relaxing monotonicity. *Biometrics* 2011; 67: 1100–1110.
56. Shepherd BE, Yu C. Accounting for data errors discovered from an audit in multiple linear regression. *Biometrics* 2011; 67: 1083–1091.
57. Koethe JR, Jenkins CA, Shepherd BE, Stinnette SE, Sterling TR. An optimal body mass index range associated with improved immune reconstitution among HIV-infected adults initiating antiretroviral therapy. *Clinical Infectious Diseases* 2011; 53: 952–960.
58. Moon TD, Burlison JR, Blevins M, Shepherd BE, Baptista A, Sidat M, Vergara AE, Vermund SH. Enrollment and programmatic trends and predictors of antiretroviral therapy initiation from President’s Emergency Plan for AIDS Relief - supported public HIV care and treatment sites in rural Mozambique. *International Journal of STD & AIDS* 2011; 22: 621–627.
59. Sahasrabuddhe VV, Bohsale RA, Kavatkar AN, Nagwanshi CA, Joshi SN, Jenkins CA, Shepherd BE, Kelkar RS, Sahay S, Risbud AR, Vermund SH, Mehendale SM. Comparison of visual inspection with acetic acid (VIA) and cervical cytology to detect high grade cervical neoplasia among HIV-infected women in India. *International Journal of Cancer* 2012; 130: 234–240.
60. Miller EK, Bugna J, Libster R, Shepherd BE, Scalzo PM, Acosta PL, Hijano D, Reynoso N, Batalle JP, Coviello S, Klein MI, Bauer G, Benitez A, Kleeberger SR, Polack FP. Human rhinoviruses in severe respiratory disease in very low birth weight infants. *Pediatrics* 2012; 129: e60–7.
61. Miller EK, Hernandez JZ, Wimmenauer V, Shepherd BE, Hijano D, Libster R, Serra ME, Bhat N, Batalle JP, Mohamed Y, Reynaldi A, Rodriguez A, Otello M, Pisa pia N, Bugna J, Bellabarba M, Kraft D, Coviello S, Ferolla FM, Chen A, London SJ, Siberry GK, Williams JV, Polack FP. A mechanistic role for type III interferon- λ_1 in asthma exacerbations mediated by human rhinoviruses. *American Journal of Respiratory and Critical Care Medicine* 2012; 185: 508–516.

62. Duda SN, Shepherd BE, Gadd CS, Masys DR, McGowan CC. Measuring the quality of observational study data in an international HIV research network. *PLoS ONE* 2012; 7: e33908.
63. Li C, Shepherd BE. A new residual for ordinal outcomes. *Biometrika* 2012; 99: 473–480.
64. Koethe JR, Blevins M, Bosire C, Nyirenda C, Kabagambe EK, Mwango A, Kasongo W, Zulu I, Shepherd BE, Heimburger DC. Self-reported dietary intake and appetite predict early treatment outcome among low body mass index adults initiating HIV treatment in sub-Saharan Africa. *Public Health Nutrition* 2012; 13:1–10.
65. Wester CW, Eden SK, Shepherd BE, Bussmann H, Novitsky V, Samuels DC, Hendrickson SL, Winkler CA, O'Brien SJ, Essex M, D'Aquila RT, Marlink RG. Risk factors for symptomatic hyperlactatemia and lactic acidosis among combination antiretroviral therapy-treated adults in Botswana: results from a clinical trial. *AIDS Research and Human Retroviruses* 2012; 28:759–65.
66. van der Heijden YF, Maruri F, Blackman A, Holt E, Warkentin JV, Shepherd BE, Sterling TR. Fluoroquinolone exposure prior to tuberculosis diagnosis is associated with an increased risk of death. *The International Journal of Tuberculosis and Lung Disease* 2012; 16: 1162–7.
67. Oswald-Richter KA, Beachboard DC, Seeley EH, Abraham S, Shepherd BE, Jenkins CA, Culver DA, Caprioli RM, Drake WP. Dual analysis for mycobacteria and propionibacteria in sarcoidosis BAL. *Journal of Clinical Immunology* 2012; 32: 1129–40.
68. Melekhin VV, Shepherd BE, Stinnette SE, Rebeiro PF, Turner MM, Sterling TR. Hemoglobin may contribute to sex differences in mortality among HIV-infected persons in care. *PLoS ONE* 2012; 7: e44999.
69. Yiannoutsos CT, Johnson LF, Boulle A, Musick BS, Gsponer T, Balestre E, Law M, Shepherd BE, Egger M. Estimated mortality of adult HIV-infected patients starting treatment with combination antiretroviral therapy. *Sexually Transmitted Infections* 2012; 88: i33-i43.
70. Audet CM, Blevins M, Moon TD, Sidat M, Shepherd BE, Pires P, Vergara A, Vermund SH. HIV/AIDS-related attitudes and practices among traditional healers in Zambezia Province, Mozambique. *Journal of Alternative and Complementary Medicine* 2012; 18:1133–41.
71. Shepherd BE, Shaw PA, Dodd LE. Using audit information to adjust parameter estimates for data errors in clinical trials. *Clinical Trials* 2012; 9: 721–729.
72. May MT, Hogg RS, Justice AC, Shepherd BE, Costagliola D, Ledergerber B, Thiebaut R, Gill MJ, Kirk O, van Sighem A, Saag M, Navarro G, Sobrino-Vegas P, Lampe F, Ingle S, Guest JL, Crane HM, D'Arminio Monforte A, Vehreschild JJ, Sterne JAC. Heterogeneity in outcomes of treated HIV-positive patients in Europe and North

- America: relation with patient and cohort characteristics. *International Journal of Epidemiology* 2012; 41: 1807–1820.
73. Parrish DD, Blevins M, Stinnette SE, Rebeiro PF, Shepherd BE, Sterling TR, McGowan CC, Wester CW. Virologic, immunologic and clinical responses in foreign-born versus US-born HIV-1 infected adults initiating antiretroviral therapy: an observational cohort study. *PLoS ONE* 2012; 7: e52336.
 74. Rasbach DA, Desruisseau AJ, Kipp AM, Stinnette S, Kheshti A, Shepherd BE, Sterling TR, Hulan T, McGowan CC, Qian HZ. Active cocaine use predicts lack of HIV-1 suppression independent of non-adherence to antiretroviral therapy: use of a rapid screening tool during routine clinic visits. *AIDS Care* 2013; 25:109–117.
 75. Koethe JR, Blevins M, Nyirenda CK, Kabagambe EK, Chiasera JM, Shepherd BE, Zulu I, Heimburger DC. Serum phosphate predicts early mortality among underweight adults starting ART in Zambia: a novel context for refeeding syndrome? *Journal of Nutrition and Metabolism* 2013; 2013:545439.
 76. Aliyu MH, Blevins M, Audet C, Shepherd BE, Hassan A, Onwujekwe O, Gebi UI, Kalish M, Lindegren ML, Vermund SH, Wester CW. Optimizing PMTCT service delivery in rural northcentral Nigeria: protocol and design for a cluster randomized study. *Contemporary Clinical Trials* 2013; 36: 187–197.
 77. Shepherd BE, Jenkins CA, Parrish DD, Glass TR, Cescon A, Masabeu A, Chene G, de Wolf F, Crane HM, Jarrin I, Gill J, del Amo J, Abgrall S, Khaykin P, Lehmann C, Ingle SM, May MT, Sterne JAC, Sterling TR; the Antiretroviral Therapy Cohort Collaboration. Higher rates of AIDS during the first year of antiretroviral therapy among migrants: the importance of tuberculosis. *AIDS* 2013; 27: 1321–1329.
 78. Mwanahamuntu MH, Sahasrabuddhe VV, Blevins M, Kapambwe S, Shepherd BE, Chibwesha C, Pfaendler KS, Mkumba G, Vwalika B, Hicks ML, Vermund SH, Stringer JSA, Parham GP. Utilization of cervical cancer screening services and test positivity rates in a ‘screen-and-treat’ cervical cancer prevention program in Zambia. *PLoS ONE* 2013; 8: e74607.
 79. Shepherd BE, Blevins M, Vaz LME, Moon TD, Kipp AM, Jose E, Ferreira FG, Vermund SH. Impact of definitions of loss to follow-up on estimates of retention, disease progression, and mortality: Application to an HIV program in Mozambique. *American Journal of Epidemiology* 2013; 178: 819–828.
 80. Gilbert PB, Shepherd BE, Hudgens MG. Sensitivity analysis of per-protocol time-to-event treatment efficacy in randomized clinical trials. *Journal of the American Statistical Association* 2013; 108: 789–800.
 81. Cortes CP, Wehbe FH, McGowan CC, Shepherd BE, Duda SN, Jenkins CA, Gonzalez E, Carriquiry G, Schechter M, Padgett D, Cesar C, Madero JS, Pape JW, Masys DR, Sterling TR. Duration of anti-tuberculosis therapy and timing of antiretroviral therapy initiation: association with mortality in HIV-related tuberculosis. *PLoS ONE* 2013; 8: e74057.

82. Tao J, Ruan Y, Yin L, Vermund SH, Shepherd BE, Shao Y, Qian HZ. Sex with Women Among Men Who Have Sex with Men in China: Prevalence and Sexual Practices. *AIDS Patient Care STDS* 2013; 27: 524–528.
83. Lu X, Mehrotra DV, Shepherd BE. Rank-based principal stratum sensitivity analyses. *Statistics in Medicine* 2013; 32: 4526–4539.
84. Huaman MA, Sterling TR, Shepherd BE, Fiske CT. 25-hydroxyvitamin D Levels after Recovery from Tuberculosis: Insights into Pathogenesis Tuberculosis. *Tuberculosis* 2014; 94: 51–54.
85. Aliyu MH, Blevins M, Parrish DD, Megazzini KM, Gebi UI, Muhammad MY, Ahmed ML, Hassan A, Shepherd BE, Vermund SH, Wester CW. Risk factors for delayed initiation of combination antiretroviral therapy: lessons from an ART program in rural north central Nigeria. *Journal of Acquired Immune Deficiency Syndromes* 2014; 65: e41–49.
86. Lou J, Blevins M, Ruan Y, Vermund SH, Tang S, Webb GF, Shepherd BE, He X, Lu H, Shao Y, Qian HZ. Modeling the impact on HIV incidence of combination prevention strategies among men who have sex with men in Beijing, China. *PLoS ONE* 2014; 9: e90985.
87. Parrish DD, Blevins M, Megazzini KM, Shepherd BE, Mohammed MY, Wester CW, Vermund SH, Aliyu MH. Hemoglobin recovery among HIV-1 infected patients on zidovudine-based antiretroviral therapy and other regimens in north-central Nigeria. *International Journal of STD & AIDS* 2014; 25: 355–359.
88. Mwanahamuntu MH, Sahasrabuddhe VV, Blevins M, Kapambwe S, Shepherd BE, Chibwesa C, Pfaendler KS, Mkumba G, Vwalika B, Hicks ML, Vermund SH, Stringer JSA, Parham GP. Monitoring the performance of screen-and-treat cervical cancer prevention programs. *International Journal of Gynecology and Obstetrics* 2014; 126: 88–89.
89. Aliyu, MH, Blevins M, Megazzini KM, Audet CM, Dunlap J, Sodangi IS, Gebi UI, Shepherd BE, Wester CW, Vermund SH. Correlates of suboptimal entry into early infant diagnosis in rural north central Nigeria. *Journal of Acquired Immune Deficiency Syndromes* 2014; 67: e19–26.
90. Boulle A, Schomaker M, May MT, Hogg RS, Shepherd BE, Monge S, Keiser O, Lampe F, Giddy J, Ndirangu J, Garone D, Fox M, Ingle SM, Reiss P, Dabis F, Costagliola D, Castagna A, Ehren K, Campbell C, Gill MJ, Saag M, Justice AC, Guest J, Crane HM, Egger M, Sterne JAC. Mortality in patients with HIV-1 infection starting antiretroviral therapy in South Africa, Europe or North America: a collaborative analysis of prospective studies. *PLoS Medicine* 2014; 11: e1001718.
91. Cesar C, Shepherd BE, Jenkins CA, Ghidinelli M, Castro JL, Valeso V, Cortes CP, Padgett D, Crabtree-Ramirez B, Gotuzzo E, Fink V, Bacon M, Sued O, McGowan CC, Cahn P. Use of third line antiretroviral therapy in Latin America. *PLoS ONE* 2014; 9: e1086887.

92. Yin L, Wang N, Vermund SH, Shepherd BE, Ruan Y, Shao Y, Qian HZ. Sexual risk reductions for HIV-infected persons: a meta-analytic review of ‘positive prevention’ randomized clinical trials. *PLoS ONE* 2014; 9: e107652.
93. Canipe A, Chidumayo T, Blevins M, Bestawros M, Bala J, Kelly P, Filteau S, Shepherd BE, Heimburger DC, Koethe JR. A 12 week longitudinal study of microbial translocation and systemic inflammation in undernourished HIV-infected Zambians initiating antiretroviral therapy. *BMC Infectious Diseases* 2014; 14:521.
94. Vermund SH, Blevins M, Moon TD, Jose E, Moiane L, Tique JA, Sidat M, Ciampa PJ, Shepherd BE, Vaz LME. Poor clinical outcomes for HIV infected children on antiretroviral therapy in rural Mozambique: need for program quality improvement and community engagement. *PLoS ONE* 2014; 9: e110116.
95. Wen SC, Tollefson SJ, Johnson M, Gilchuk P, Boyd KL, Shepherd BE, Joyce S, Williams JV. Acute clearance of human metapneumovirus occurs independently of natural killer cells. *Journal of Virology* 2014; 88: 10963–10969.
96. Koethe JR, Jenkins CA, Turner M, Bebawy SS, Wester CW, Shepherd BE, Sterling TR. Body mass index and the risk of incident non-communicable diseases after starting antiretroviral therapy. *HIV Medicine* 2015; 16: 67–72.
97. Ray WA, Liu Q, Shepherd BE. Performance of time-dependent propensity scores: a pharmacoepidemiology case study. *Pharmacoepidemiology and Drug Safety* 2015; 24: 98–106.
98. Blevins M, Jose E, Bilhete FR, Vaz LME, Shepherd BE, Audet CM, Vermund SH, Moon TD. Two-year death and loss to follow-up outcomes by source of referral to HIV care for HIV-infected patients initiating antiretroviral therapy in rural Mozambique. *AIDS Research and Human Retroviruses* 2015; 31: 198–207.
99. Castilho JL, Luz PM, Shepherd BE, Turner M, Rebeiro SR, Bebawy SS, Netto JS, McGowan CC, Veloso VG, Engels EA, Sterling TR, Grinsztejn B. HIV and cancer: a comparative retrospective study of Brazilian and U.S. clinical cohorts. *Infectious Agents and Cancer* 2015; 10: 4.
100. Aliyu MH, Blevins M, Arinze F, Megazzini KM, Bussell, Dunlap J, Odoh C, Gebi UI, Muhammad MY, Shepherd BE, Audet CM, Vermund SH, Wester CW. Enrolment trends in a comprehensive HIV programme in rural north-central Nigeria: improved care indices, but declining quality of clinical data over time. *Pathogens and Global Health* 2015; 109: 75–83.
101. Bestawros M, Chidumayo T, Blevins M, Canipe A, Bala J, Kelly P, Filteau S, Shepherd BE, Heimburger DC, Koethe JR. Increased systemic inflammation is associated with cardiac and vascular dysfunction over the first 12 weeks of antiretroviral therapy among undernourished, HIV-infected adults in southern Africa. *Journal of AIDS and Clinical Research* 2015; 6: 3.

102. *Giganti MJ, Luz PM, Caro-Vega Y, Cesar C, Padgett D, Koenig S, Echevarria J, McGowan CC, Shepherd BE. A comparison of seven Cox regression-based models to account for heterogeneity across multiple HIV treatment cohorts in Latin America and the Caribbean. *AIDS Research and Human Retroviruses* 2015; 31: 496–503.
103. Audet CM, Gutin SA, Blevins M, Chiau E, Alvim F, Jose E, Vaz LME, Shepherd BE, Dawson-Rose C. The impact of visual aids and enhanced training on the delivery of Positive Health, Dignity, and Prevention messages to adult patients living with HIV in rural north central Mozambique. *PLoS ONE* 2015; 10:e0130676.
104. Carriquiry G, Fink V, Koethe JR, Giganti MJ, Jayathilake K, Blevins M, Cahn P, Grinsztejn B, Wolff M, Pape JW, Padgett D, Madero JS, Gotuzzo E, McGowan CC, Shepherd BE. Mortality and loss to follow-up among HIV-infected persons on long-term antiretroviral therapy in Latin America and the Caribbean. *Journal of the International AIDS Society* 2015; 18: 20016.
105. Luz PM, Belaunzaran-Zamudio PF, Crabtree-Ramirez B, Caro-Vega Y, Hoces D, Rebeiro PF, Blevins M, Pape JW, Cortes CP, Padgett D, Cahn P, Veloso VG, McGowan CC, Grinsztejn B, Shepherd BE. CD4 response up to 5 years following combination antiretroviral therapy in HIV-infected patients in Latin America and the Caribbean. *Open Forum Infectious Diseases* 2015; 2:ofv079.
106. da Silva M, Blevins M, Wester CW, Manjolo J, Jose E, Gonzalez LC, Shepherd BE, Moon TD, Vaz LME. Patient loss to follow-up before antiretroviral therapy initiation in rural Mozambique. *AIDS and Behavior* 2015; 19: 666–678.
107. Kapambwe S, Sahasrabuddhe VV, Blevins M, Mwanahamuntu MH, Mudenda V, Shepherd BE, Chibwesha C, Pfaendler KS, Hicks ML, Vermund SH, Stringer JSA, Parham GP. Age distribution and determinants of invasive cervical cancer in a 'screen-and-treat' program integrated with HIV/AIDS care in Zambia. *Journal of Acquired Immune Deficiency Syndromes* 2015; 70: e20–26.
108. *Wanga V, Venuto C, Morse GD, Acosta EP, Daar ES, Haas DW, Li C, Shepherd BE. Genome-wide Association Study of Tenofovir Pharmacokinetics and Creatinine Clearance in AIDS Clinical Trials Group Protocol A5202. *Pharmacogenetics and Genomics* 2015; 25: 450–461.
109. Koethe JR, Jenkins CA, Lau B, Shepherd BE, Silverberg MJ, Brown TT, Blashill AJ, Anema A, Willig A, Stinnette S, Napravnik S, Gill J, Crane HM, Sterling TR. Body mass index and early CD4 T-cell recovery among adults initiating antiretroviral therapy in North America, 1998-2010. *HIV Medicine* 2015; 16: 572–577.
110. Castilho JL, Jenkins CA, Shepherd BE, Bebawy SS, Turner M, Sterling TR, Melekhin VV. Hormonal contraception and risk of psychiatric and other non-communicable diseases in HIV-infected women. *Journal of Women's Health* 2015; 24: 481–488.
111. Crabtree-Ramirez B, Vega YNC, Shepherd BE, Turner M, Carriquiry G, Fink V, Luz PM, Cortes CP, Rouzier V, Padgett D, Jayathilake K, McGowan CC, Person

- AK. Temporal trends in age at HIV diagnosis in cohorts in the United States, the Caribbean, and Central and South America. *AIDS and Behavior* 2015; 19: 1599–1608.
112. Huaman MA, Fiske CT, Jones TF, Warkentin J, Shepherd BE, Ingram LA, Maruri F, Sterling TR. Tuberculosis and the risk of infection with other intracellular bacteria: a population-based study. *Epidemiology and Infection* 2015; 143: 951–959.
 113. Vandenhende MA, Ingle S, May M, Chene G, Zangerle R, Van Sighems A, Gill MJ, Schwarze-Zander C, Hernandez-Novoa B, Obel N, Kirk O, Abgrall S, Guest J, Samji H, D’Arminio Monforte A, Llibre JM, Smith C, Cavassini M, Burkholder GA, Shepherd B, Crane HM, Sterne J, Morlat P. Impact of low-level viremia on clinical and virological outcomes in treated HIV-1 Infected Patients. *AIDS* 2015; 29: 373–383.
 114. Belaunzaran-Zamudio PF, Caro-Vega YN, Shepherd BE, Crabtree-Ramirez BE, Luz PM, Grinsztejn B, Cesar C, Cahn P, Cortes C, Wolff M, Pape JW, Padgett D, Gotuzzo E, McGowan CC, Sierra-Madero JG. Monitoring of HIV treatment in seven countries in WHO region of the Americas. *Bulletin of the World Health Organization* 2015; 93: 529-539.
 115. Aliyu MH, Blevins M, Megazzini KM, Parrish DD, Audet CM, Chan N, Odoh C, Gebi UI, Muhammad MY, Shepherd BE, Wester CW, Vermund SH. Pregnant women with HIV in rural Nigeria have higher rates of antiretroviral treatment initiation, but similar loss to follow-up as non-pregnant women and men. *International Health* 2015; 7:405–411.
 116. Cesar C, Jenkins CA, Shepherd BE, Padgett D, Mejia F, Rebeiro SR, Cortes C, Pape JW, Madero JS, Fink V, Sued O, McGowan CC, Cahn P. Incidence of virological failure and major regimen change of initial combination antiretroviral therapy in Latin America and the Caribbean: an observational cohort study. *Lancet HIV* 2015; 2: e492–500.
 117. Koethe JR, Jenkins CA, Lau B, Shepherd BE, Justice AC, Tate JP, Buchacz K, Napravnik S, Mayor AM, Horberg MA, Blashill AJ, Willig A, Wester CW, Silverberg MJ, Gill J, Thorne JE, Klein M, Eron JJ, Kitahata MM, Sterling TR, Moore RD. Rising obesity prevalence and weight gain among adults starting antiretroviral therapy in the United States and Canada. *AIDS Research and Human Retroviruses* 2016; 32:50–58.
 118. Kipp AM, Blevins M, Haley CA, Mwinga K, Habimana P, Shepherd BE, Aliyu MH, Ketsela T, Vermund SH. Factors associated with declining under-five mortality rates from 2000 to 2013: an ecological analysis of 46 African countries. *British Medical Journal Open* 2016; 6:e007675.
 119. Wolff M, Shepherd BE, Cortes C, Rebeiro P, Cesar C, Cardoso SW, Pape JW, Padgett D, Madero JS, Echevarria J, McGowan CC. Clinical and virologic outcomes after changes in first antiretroviral regimen at 7 sites in the Caribbean, Central

- and South America Network (CCASAnet). *Journal of Acquired Immune Deficiency Syndromes* 2016; 71:102–110.
120. Qian HZ, Ruan Y, Liu Y, Milam DF, Spiegel HML, Yin L, Li D, Shepherd BE, Shao Y, Vermund SH. Lower HIV risk among circumcised men who have sex with men in China: Interaction with anal sex role in a cross-sectional study. *Journal of Acquired Immune Deficiency Syndromes* 2016; 71:444–451.
 121. Blevins M, Wehbe FH, Rebeiro PF, Caro-Vega Y, McGowan CC, Shepherd BE. Interactive data visualization for HIV cohorts: leveraging data exchange standards to share and reuse research tools. *PLoS ONE* 2016; 11: e0151201.
 122. Cesar C, Koethe JR, Giganti MJ, Rebeiro P, Althoff KN, Napravnik S, Mayor A, Grinsztejn B, Wolff M, Padgett D, Sierra-Madero J, Gotuzzo E, Sterling TR, Willig J, Levison J, Kitahata M, Rodriguez-Barradas MC, Moore RD, McGowan CC, Shepherd BE, Cahn P for CCASAnet and NA-ACCORD. Health outcomes among HIV-infected Latinos initiating antiretroviral therapy in North America versus Central and South America. *Journal of the International AIDS Society* 2016; 19: 20684.
 123. Castilho JL, Shepherd BE, Koethe JR, Turner M, Bebawy S, Logan J, Rogers WB, Raffanti S, Sterling TR. CD4/CD8 ratio, age, and risk of serious non-communicable diseases in HIV-infected adults on antiretroviral therapy. *AIDS* 2016; 30: 899–908.
 124. Aliyu MH, Blevins M, Audet CM, Kalish M, Gebi UI, Onwujekwe O, Lindegren ML, Shepherd BE, Wester CW, Vermund SH. Integrated prevention of mother-to-child HIV transmission services, antiretroviral therapy initiation, and maternal and infant retention in care in rural north-central Nigeria: a cluster-randomised controlled trial. *Lancet HIV* 2016; 3: e202–211.
 125. Rebeiro PF, Cesar C, Shepherd BE, De Boni RB, Cortes CP, Rodriguez F, Belaunzaran-Zamudio P, Pape JW, Padgett D, Hoces D, McGowan CC, Cahn P. Assessing the HIV care continuum in Latin America: progress in clinical retention, cART use, and viral suppression. *Journal of the International AIDS Society* 2016; 19: 20636.
 126. Koethe JR, Jenkins CA, Petucci C, Culver J, Shepherd BE, Sterling TR. Superior glucose tolerance and metabolomic profiles, independent of adiposity, in HIV-infected women compared to men on antiretroviral therapy. *Medicine* 2016; 95: e3634.
 127. Crabtree-Ramirez B, Caro-Vega Y, Shepherd BE, Grinsztejn B, Wolff M, Cortes CP, Padgett D, Carriquiry G, Fink V, Jayathilake K, Person A, McGowan C, Sierra-Madero J. Time to HAART initiation after diagnosis and treatment of opportunistic infections in patients with AIDS in Latin America. *PLoS ONE* 2016; 11: e0153921.
 128. Koethe JR, Jenkins CA, Lau B, Shepherd BE, Wester W, Rebeiro PF, Silverberg MJ, Thorne JE, Gill J, Mayor AM, Willig A, Bosch R, Horberg MA, Justice AC, Sterling TR, Moore RD. Higher time-updated body mass index: association with

- improved CD4+ cell recovery on HIV treatment. *Journal of Acquired Immune Deficiency Syndromes* 2016; 73: 197–204.
129. Pettit AC, Mendes A, Jenkins C, Napravnik S, Freeman A, Shepherd BE, Dowdy D, Gill J, Rachlis A, Moore R, Sterling TR. Timing of antiretroviral treatment, immunovirologic status and TB risk: implications for test and treat. *Journal of Acquired Immune Deficiency Syndromes* 2016; 72:572–578.
 130. De Boni R, Shepherd BE, Grinsztejn B, Cesar C, Cortes C, Padgett D, Gotuzzo E, Belaunzaran-Zamudio F, Rebeiro PF, Duda SN, McGowan CC. Substance use and adherence among people living with HIV/AIDS receiving cART in Latin America. *AIDS and Behavior* 2016; 20: 2692–2699.
 131. ⁺ Shepherd BE, Liu Q, Mercaldo N, Jenkins CA, Lau B, Cole SR, Saag MS, Sterling TR. Comparing results from multiple imputation and dynamic marginal structural models for estimating when to start antiretroviral therapy. *Statistics in Medicine* 2016; 35: 4335–4351.
 132. ^{**} Shepherd BE, Li C, Liu Q. Probability-scale residuals for continuous, discrete, and censored data. *Canadian Journal of Statistics* 2016; 44: 463–479.
 133. Wester C, Rebeiro PF, Shavor TJ, Shepherd BE, McGoy S, Daley B, Morrison M, Vermund SH, Pettit AC. The 2013 HIV continuum of care in Tennessee: progress despite persistent disparities. *Public Health Reports* 2016; 131: 695–703.
 134. Cain LE, Saag MS, Petersen M, May MT, Ingle SM, Logan R, Robins JM, Abgrall S, Shepherd BE, Deeks SG, Gill MJ, Touloumi G, Vourli G, Dabis F, Vandenhede M, Reiss P, van Sighem A, Samji H, Hogg R, Rybniker J, Sabin CA, Jose S, del Amo J, Moreno S, Rodriguez B, Cozzi-Lepri A, Boswell SL, Stephan C, Perez-Hoyos S, Jarrin I, Guest JL, Monforte AD, Antinori A, Moore R, Esteve A, Casabona J, Meyer L, Seng R, Phillips A, Bucher HC, Egger M, Mugavero M, Haubrich R, Geng E, Olson A, Eron JJ, Napravnik S, Kitahata MM, Van Rompaey SE, Teira R, Justice AC, Tate J, Costagliola D, Sterne JA, Hernan MA. Using observational data to emulate a randomized trial of dynamic treatment switching strategies: an application to antiretroviral therapy. *International Journal of Epidemiology* 2016; 45: 2038–2049.
 135. Tao J, Qian HZ, Kipp AM, Ruan Y, Shepherd BE, Amico KR, Shao Y, Lu H, Vermund SH. Effects of depression and anxiety on antiretroviral therapy adherence among newly diagnosed HIV-infected Chinese men who have sex with men. *AIDS* 2017; 31: 401–406.
 136. Tao J, Vermund SH, Lu H, Ruan Y, Shepherd BE, Kipp AM, Amico KR, Zhang X, Shao Y, Qian HZ. Impact of depression and anxiety on initiation of antiretroviral therapy among men who have sex with men with newly diagnosed HIV infections in China. *AIDS Patient Care and STDs* 2017; 31: 96–104.

137. Shepherd BE, Rebeiro PF. Assessing and interpreting the association between continuous covariates and outcomes in observational studies of HIV using splines. *Journal of Acquired Immune Deficiency Syndromes* 2017; 74: e60–e63.
138. Luque MT, Jenkins CA, Shepherd BE, Padgett D, Rouzier V, Succi RCM, Machado DM, McGowan CC, Vermund SH, Pinto JA. Mortality in children with human immunodeficiency virus initiating treatment: a six-cohort study in Latin America. *Journal of Pediatrics* 2017; 182: 245–252.
139. Anderegg N, Johnson LF, Zaniwski E, Althoff K, Balestre E, Law M, Nash D, Shepherd BE, Yiannoutsos CT, Egger M. All-cause mortality in HIV-positive adults starting combination antiretroviral therapy: correcting for loss to follow-up. *AIDS* 2017; 31: S31–S40.
140. van der Heijden YF, Karim F, Mufamadi G, Zako L, Chinappa T, Shepherd BE, Maruri F, Moosa Y, Sterling TR, Pym AS. Isoniazid mono-resistant tuberculosis is associated with poor treatment outcomes in Durban, South Africa. *The International Journal of Tuberculosis and Lung Disease* 2017; 21: 670–676.
141. Wolff MJ, Giganti MJ, Cortes CP, Cahn P, Grinsztejn B, Pape JW, Padgett D, Sierra-Madero J, Gotuzzo E, Duda SN, McGowan CC, Shepherd BE. A decade of HAART in Latin America: a longitudinal continuum of care among the first wave of HIV patients to receive antiretroviral. *PLoS ONE* 2017; 12: e0179769.
142. Kipp AM, Rebeiro PF, Shepherd BE, Brinkley-Rubinstein L, Turner M, Bebawy S, Sterling TR, Hulgán T. Daily marijuana use is associated with missed clinic appointments among HIV-infected persons engaged in HIV care. *AIDS and Behavior* 2017; 21: 1996–2004.
143. Shepherd BE, Blevins Peratikos M, Rebeiro PF, Duda SN, McGowan CC. A pragmatic approach for reproducible research with sensitive data. *American Journal of Epidemiology* 2017; 186: 387–392.
144. Crabtree Ramirez B, Caro Vega Y, Shepherd BE, Le C, Turner M, Cahn P, Grinsztejn B, Cortes C, Padgett D, Sterling TR, McGowan CC, Person A. Outcomes of HIV-positive patients with cryptococcal meningitis in the Americas. *International Journal of Infectious Diseases* 2017; 63: 57–63.
145. ⁺ *Liu Q, Shepherd BE, Li C, Harrell FE. Modeling continuous response variables using ordinal regression. *Statistics in Medicine* 2017; 36: 4316–4335.
146. Norwood J, Koethe JR, Turner M, Bofill C, Rebeiro P, Shepherd BE, Bebawy S, Hulgán T, Raffanti S, Haas DW, Sterling TR. Weight gain in persons with HIV switched from efavirenz-based to integrase strand transfer inhibitor-based regimens. *Journal of Acquired Immune Deficiency Syndromes* 2017; 76: 527–531.
147. De Boni RB, Peratikos MB, Shepherd BE, Grinsztejn B, Cortes C, Padgett D, Gotuzzo E, Belaunzaran-Zamudio PF, Rebeiro PF, Duda SN, McGowan CC. Is

- substance use associated with HIV cascade outcomes in Latin America? *PLoS ONE* 2018; 13: e0194228.
148. Oh EJ, Shepherd BE, Lumley T, Shaw PA. Considerations for analysis of time-to-event outcomes measured with error: bias and correction with SIMEX. *Statistics in Medicine* 2018; 37: 1276–1289.
 149. Pettit AC, Giganti MJ, Ingle SM, May MT, Shepherd BE, Gill MJ, Fatkenheuer G, Abgrall S, Saag MS, Del Amo J, Justice AC, Miro JM, Cavasinni M, Dabis F, Monforte AD, Reiss P, Guest J, Moore D, Shepherd L, Obe N, Crane HM, Smith C, Teira R, Zangerle R, Sterne JAC, Sterling TR. Increased non-AIDS mortality among persons with AIDS defining events after antiretroviral therapy initiation. *Journal of the International AIDS Society* 2018; 21: e25031.
 150. Liu Y, Vermund SH, Ruan Y, Liu H, Amico KR, Simoni JM, Shepherd BE, Shao Y, Qian HZ. Peer counseling versus standard-of-care on reducing high-risk behaviors among newly diagnosed HIV-positive men who have sex with men in Beijing, China: a randomized intervention study. *Journal of the International AIDS Society* 2018; 21: e25079.
 151. Carriquiry G, Giganti MJ, Castilho JL, Jayathilake K, Cahn P, Grinsztejn B, Cortes C, Pape JW, Padgett D, Madero JS, McGowan CC, Shepherd BE, Gotuzzo E. Virologic Failure and Mortality in Older ART Initiators in a Multisite Latin American and Caribbean Cohort. *Journal of the International AIDS Society* 2018; 21: e25088.
 152. Caro-Vega Y, Belaunzaran-Zamudio PF, Crabtree-Ramirez BE, Shepherd BE, Grinsztejn B, Wolff M, Pape JW, Padgett D, Gotuzzo E, McGowan CC, Sierra-Madero JG. Durability of efavirenz compared with boosted protease inhibitor-based regimens in antiretroviral-naïve patients in the Caribbean and Central and South America. *Open Forum Infectious Diseases* 2018; 5: ofy004.
 153. Fink VI, Jenkins CA, Castilho J, Person AK, Shepherd BE, Grinsztejn B, Netto J, Crabtree B, Cortes CP, Padgett D, Jayathilake K, McGowan CC, Cahn P. Survival after cancer diagnosis in a cohort of HIV-positive individuals in Latin America. *Infectious Agents and Cancer* 2018; 13:16.
 154. *Liu Q, Li C, Wanga V, Shepherd BE. Covariate-adjusted Spearman’s rank correlation with probability-scale residuals. *Biometrics* 2018; 74: 595–605.
 155. Bailin S, Jenkins C, Petucci C, Culver JA, Shepherd BE, Fessel JP, Hulgán T, Koethe JR. Lower concentrations of circulating medium and long chain acylcarnitines characterize insulin resistance in persons with HIV. *AIDS Research and Human Retroviruses* 2018; 34: 536–543.
 156. Caro-Vega Y, Belaunzaran-Zamudio PF, Crabtree-Ramirez B, Shepherd BE, Mejia F, Giganti MJ, Patterson P, Grinsztejn B, Wolff M, Pape JW, Padgett D, Castilho JL, McGowan CC, Sierra-Madero JG. Trends in proportion of older HIV-infected people in care in Latin America and the Caribbean: A growing challenge. *Epidemiology and Infection* 2018; 146: 1308–1311.

157. Setliff I, McDonnell WJ, Raju N, Bombardi RG, Murji AA, Scheepers C, Ziki R, Mynhardt C, Shepherd BE, Mamchak AA, Garrett N, Karim SA, Mallal SA, Crowe JE Jr, Morris L, Georgiev IS. Multi-donor longitudinal antibody repertoire sequencing reveals the existence of public antibody clonotypes in HIV-1 infection. *Cell Host and Microbe* 2018; 23: 845–854.
158. Audet CM, Graves E, Barreto E, De Schacht C, Gong, Shepherd BE, Aboobacar A, Calvo L, Alvim MF, Aliyu MH, Kipp Am, Jordan H, Amico R, Diemer M, Ciaranello A, Dugdale C, Vermund SH, Rompaey SV. Partners-based HIV treatment for seroconcordant couples attending antenatal and postnatal care in rural Mozambique: a cluster randomized trial protocol. *Contemporary Clinical Trials* 2018; 71: 63–69.
159. van der Heijden YF, Karim F, Chinappa T, Mufamadi G, Zako L, Shepherd BE, Maruri F, Moosa MS, Sterling TR, Pym AS. Older age at first tuberculosis diagnosis is associated with tuberculosis recurrence in HIV-negative persons. *International Journal of Tuberculosis and Lung Disease* 2018; 22: 871–877.
160. Coelho L, Rebeiro PF, Castilho JL, Caro-Veja Y, Mejia FA, Cesar C, Cortes CP, Padgett D, McGowan CC, Veloso VG, Sterling TR, Brinsztejn B, Shepherd BE, Luz PM. Early retention in care neither mediates nor modifies the effect of sex and sexual mode of HIV acquisition on HIV survival in the Americas. *AIDS Patient Care and STDs* 2018; 32: 306–313.
161. Koethe JR, Jenkins CA, Furch BD, Lake JE, Barnett L, Hager CC, Smith R, Hulgán T, Shepherd BE, Kalams SA. Circulating markers of immunologic activity reflect adiposity in persons with HIV on antiretroviral therapy. *Journal of Acquired Immune Deficiency Syndromes* 2018; 79: 135–140.
162. Pettit AC, Shepherd BE, Sterling TR. Treatment of drug-susceptible tuberculosis among people living with Human Immunodeficiency Virus infection: an update. *Current Opinion in HIV and AIDS* 2018; 13: 469–477.
163. Crabtree-Ramirez BE, Jenkins C, Jayathilake K, Carriquiry G, Veloso VG, Padgett D, Gotuzzo E, Cortes C, Mejia F, McGowan CC, Duda S, Shepherd BE, Sterling TR. HIV-related tuberculosis: mortality risk in persons without vs. culture-confirmed disease. *International Journal of Tuberculosis and Lung Disease* 2019; 23: 306–314.
164. Adcox JD, Shepherd BE, Messersmith M. An analysis of image quality and ease of use for six orthodontic camera systems. *Journal of Clinical Orthodontics* 2019; 53: 139–148.
165. Aliyu MH, Wudil UJ, Ingles DJ, Shepherd BE, Gong W, Musa BM, Muhammad H, Mahmoud SU, Abdu A, Atanda A, Ahonkhai AA, Ikizler TA, Winkler CA, Kopp JB, Kimmel PL, Wester CW. Optimal management of HIV-infected adults at risk for kidney disease in Nigeria (Renal Risk Reduction "R3" Trial): protocol and study design. *Trials* 2019; 20: 341.

166. Rebeiro PF, Pettit C, Sizemore L, Mathieson SA, Wester C, Kipp A, Shepherd BE, Sterling TR. Trends and disparities in mortality and progression to AIDS in the highly active antiretroviral therapy era, Tennessee 1996-2016. *American Journal of Public Health* 2019; 109: 1266–1272.
167. Castilho JL, Turner M, Shepherd BE, Koethe JR, Furukawa SS, Bofill CE, Raf-fanti S, Sterling TR. CD4/CD8 ratio and CD4 nadir predict mortality following non-communicable disease diagnosis in adults living with HIV. *AIDS Research and Human Retroviruses* 2019; 35: 960–967.
168. Pettit AC, Jenkins CA, Peratikos MB, Yotebieng M, Diero L, Do CD, Ross J, Veloso VG, Hawerlander D, Marcy O, Shepherd BE, Fenner L, Sterling TR. Directly Ob-served Therapy and Risk of Unfavorable Tuberculosis Treatment Outcomes Among an International Cohort of People Living with HIV in and Low-and Middle-Income Countries. *Journal of the International AIDS Society* 2019; 22: e25423.
169. Crabtree-Ramirez BE, Caro-Vega YN, Belaunzaran-Zamudio PF, Shepherd BE, Re-beiro PF, Veloso V, Cortes CP, Padgett D, Gotuzzo E, Sierra-Madero J, McGowan CC, Person AK. Temporal Changes in ART Initiation in Adults with High CD4 Counts in Latin America. *Journal of the International AIDS Society* 2019; 22: e25413.
170. *Giganti MJ, Shepherd BE, Caro-Vega Y, Luz PM, Rebeiro PF, Maia M, Julmiste G, Cortes C, McGowan CC, Duda SN. The impact of data quality and source data verification on epidemiologic inference: a practical application using HIV observa-tional data. *BMC Public Health* 2019; 19: 1748.
171. *Lotspeich S, Giganti M, Maia M, Vieira R, Machado D, de Menezes Succi R, Ribeiro S, Pereira M, Rodriguez M, Julmiste G, Luque M, Caro-Vega Y, Mejia F, Shepherd BE, McGowan CC, Duda S. Self-audits as alternatives to travel-audits for improving data quality in the Caribbean, Central and South America network for HIV epidemiology. *Journal of Clinical and Translational Science* 2019; 4: 125–132.
172. Oliver CD, Rebeiro PF, Shepherd BE, Keruly J, Mayer KH, Mathews WC, Turan B, Moore RD, Crane HM, Geng E, Napravnik S, Kitahata MM, Mugavero MJ, Pettit AC. Clinic-level factors associated with retention in care among people living with HIV in a multi-site United States cohort, 2010-2016. *Clinical Infectious Diseases* 2020; 71: 2592–8.
173. Zhang C, Webb GF, Lou J, Shepherd BE, Qian HZ, Liu Y, Vermund SH. Predicting the long-term impact of voluntary medical male circumcision on HIV incidence among men who have sex with men in Beijing, China. *AIDS Care* 2020; 32: 343–353.
174. *Tian Y, Hothorn T, Li C, Harrell FE, Shepherd BE. An empirical comparison of two novel transformation models. *Statistics in Medicine* 2020; 39: 562–576.

175. Castilho JL, Rebeiro PF, Shepherd BE, Nash R, Adams RS, Turner M, Furukawa S, Hulgán T, Koethe JR, Sterling TR. Mood disorders and increased risk of non-communicable disease in adults with HIV. *Journal of Acquired Immune Deficiency Syndromes* 2020; 83: 397–404.
176. Lee JT, Saag L, Kipp AM, Logan J, Shepherd BE, Koethe JR, Turner M, Bebawy SS, Sterling TR, Hulgán T. Self-reported Cannabis Use and Changes in Body Mass Index, CD4 T-cell Counts, and HIV-1 RNA Suppression in Treated Persons with HIV. *AIDS in Behavior* 2020; 24: 1275–1280.
177. Bourgi K, Jenkins CA, Rebeiro PF, Shepherd BE, Palella F, Moore RD, Althoff KN, Gill J, Rabkin CS, Gange SJ, Horberg MA, Margolick J, Li J, Wong C, Willig A, Lima V, Crane H, Thorne J, Silverberg MJ, Kirk G, Mathews WC, Sterling TR, Lake J, Koethe JR. Weight gain among treatment-naïve persons with HIV starting integrase inhibitors compared to non-nucleoside reverse transcriptase inhibitors or protease inhibitors in a large observational cohort in the United States and Canada. *Journal of the International AIDS Society* 2020; 23: 25484.
178. Belaunzaran-Zamudio PF, Caro-Vega Y, Giganti MJ, Castilho JL, Crabtree-Ramirez BE, Shepherd BE, Mejia F, Cesar C, Moreira RC, Wolff M, Pape JW, Padgett D, McGowan CC, Sierra-Madero JG. Frequency of non-communicable diseases in people 50 years of age and older receiving HIV care in Latin America. *PLoS ONE* 2020; 15: e0233965.
179. *Giganti MJ, Shaw PA, Chen G, Bebawy SS, Turner MM, Sterling TR, Shepherd BE. Accounting for dependent errors in predictors and time-to-event outcomes using electronic health record, validation samples, and multiple imputation. *Annals of Applied Statistics* 2020; 14: 1045–1061.
180. Gilbert PB, Blette BS, Shepherd BE, Hudgens MG. Post-randomization biomarker effect modification analysis in an HIV vaccine clinical trial. *Journal of Causal Inference* 2020; 8: 54–69.
181. Belaunzaran-Zamudio PF, Caro-Vega YN, Shepherd BE, Rebeiro PF, Crabtree-Ramirez BE, Cortes CP, Grinsztejn B, Gotuzzo E, Mejia F, Padgett D, Pape JW, Rouzier V, Veloso V, Cardoso SW, McGowan CC, Sierra-Madero JG. The population impact of late presentation with advanced HIV disease and delayed antiretroviral therapy in adults receiving HIV care in Latin America. *American Journal of Epidemiology* 2020; 189: 564–572.
182. Koenig SP, Kim A, Shepherd BE, Cesar C, Veloso V, Cortes CP, Padgett D, Crabtree-Ramirez B, Gotuzzo E, McGowan CC, Sterling TR, Pape JW. Increased Mortality after Tuberculosis Treatment Completion in Persons with HIV in Latin America. *Clinical Infectious Diseases* 2020; 71: 215–217.
183. *Liu Q, Shepherd BE, Li C. PResiduals: an R package for residual analysis using probability-scale residuals. *Journal of Statistical Software* 2020; 94: 12. doi: 10.18637/jss.v094.i12

184. Were MC, Gong W, Balirwa P, Balugaba BE, Yeung A, Pierce L, Ingles D, Kim Y, Shepherd BE. Coverage of IMIA-recommended competencies by Masters in Health Informatics degree programs in East Africa. *International Journal of Medical Informatics* 2020; 143: 104265.
185. *Giganti MJ, Shepherd BE. Multiple imputation variance estimation in studies with missing or misclassified inclusion criteria. *American Journal of Epidemiology* 2020; 189: 1628–1632.
186. Friebe MP, LeGrand JR, Shepherd BE, Breeden B, Nelson SD. Reducing inappropriate outpatient medication prescribing in older adults across EHR systems. *Applied Clinical Informatics* 2020; 11: 865–872.
187. Shepherd BE, Shaw PA. Errors in multiple variables in HIV cohort and electronic health record data: statistical challenges and opportunities. *Statistical Communications in Infectious Diseases* 2020; 12: 20190015.
188. Castilho JL, Kim A, Jenkins CA, Grinsztejn B, Gotuzzo E, Fink V, Padgett D, Belaunzaran-Zamudio PF, Crabtree-Ramirez B, Escuder MM, Souza RA, Tenore SB, Pimentel SR, Ideda MLR, de Alencastro PR, Tupinanbas U, Brites C, Luz E, Netto J, Cortes CP, Grangeiro A, Shepherd BE, McGowan CC. Antiretroviral therapy and Kaposi’s sarcoma trends and outcomes among adults with HIV in Latin America. *Journal of the International AIDS Society* 2021; 24: e25658.
189. Shaw PA, He J, Shepherd BE. Regression calibration to correct correlated errors in outcome and exposure. *Statistics in Medicine* 2021; 40: 271–286.
190. Tao R, Lotspeich SC, Amorim G, Shaw PA, Shepherd BE. Efficient semiparametric inference for two-phase studies with outcome and covariate measurement errors. *Statistics in Medicine* 2021; 40: 725–738.
191. Oh EJ, Shepherd BE, Lumley T, Shaw PA. Raking and regression calibration: methods to address bias from correlated covariate and time-to-event error. *Statistics in Medicine* 2021; 40: 631–649.
192. Pereira GFM, Kim A, Jalil EM, Fonseca FF, Shepherd BE, Veloso VG, Rick F, Ribeiro R, Pimenta MC, Beber A, Correa RG, Lima R, Maruri F, McGowan CC, Benzaken A, Grinsztejn B, Castilho JL. Dolutegravir and pregnancy outcomes among women on ART in Brazil: a national cohort study. *Lancet HIV* 2021; 8: e33–e41.
193. Cooper RL, Brown LL, Tabatabai M, Shepherd BE, Haas DW, Myers HF, Edgerton RD, Bonny C, Watson JA, Berthaud V. The effects of perceived stress and cortisol concentration on antiretroviral adherence when mediated by psychological flexibility among southern black men living with HIV. *AIDS and Behavior* 2021; 25: 645–652.
194. Audet CM, Shepherd BE, Aliyu MH, Moshabela M, Pettapiece-Phillips MJ, Wagner RG. Healer led vs. clinician led training to improve personal protective equipment

- use among traditional healers in South Africa: a randomized controlled trial protocol. *Global Health Action* 2021; 14: 1, 1898131.
195. Han K, Lumley T, Shepherd BE, Shaw PA. Two-phase analysis and study design for survival models with error-prone exposures. *Statistical Methods in Medical Research* 2021; 30: 857–874.
 196. Oh EJ, Shepherd BE, Lumley T, Shaw PA. Improved generalized raking estimators to address dependent covariate and failure-time outcome error. *Biometrical Journal* 2021; 63: 1006–1027.
 197. Ahonkhai AA, Aliyu MH, Audet CM, Bravo M, Simmons M, Claquin G, Memiah P, Fernando AN, Carlucci JG, Shepherd BE, Van Rompaey S, Yu Z, Gong W, Vermund SH, Wester CW. Poor retention and care-related sex disparities among youth living with HIV in rural Mozambique. *PLoS ONE* 2021; 16: e0250921.
 198. Yohannes NT, Jenkins CA, Clouse K, Cortes CP, Cordero FM, Padgett D, Rouzier V, Friedman RK, McGowan CC, Shepherd BE, Rebeiro PF. Timing of HIV diagnosis relative to pregnancy and postpartum HIV care continuum outcomes among Latin American women, 2000 to 2017. *Journal of the International AIDS Society* 2021; 24: e25740.
 199. Somerville K, Jenkins CA, Carlucci JG, Person AK, Machado DM, Luque MT, Pinto JA, Rouzier V, Friedman RK, McGowan CC, Shepherd BE, Rebeiro PF. Outcomes after second-line antiretroviral therapy in children living with HIV in Latin America. *Journal of Acquired Immune Deficiency Syndromes* 2021; 87: 993–1001.
 200. Were MC, Gong W, Balirwa P, Balugaba BE, Yeung A, Pierce L, Ingles D, Kim Y, Shepherd BE. Comparative analysis of competency coverage within accredited master’s in health informatics programs in the East African region. *Journal of the American Medical Informatics Association* 2021; 28: 1843–1848.
 201. Patel R, Amorim G, Jakait B, Shepherd BE, Mocello R, Musick B, Bernard C, Onono M, Bukusi E, Wools-Kaloustian K, Cohen C, Yiannoutsos C. Pregnancies among women living with HIV using contraceptives and antiretroviral therapy in western Kenya: a retrospective cohort study. *BMC Medicine* 2021; 13: 178.
 202. Pettit AC, Bian A, Schember CO, Rebeiro PF, Keruly JC, Mayer KH, Mathews WC, Moore RD, Crane HM, Geng E, Napravnik S, Shepherd BE, Mugavero MJ. Development and validation of a multivariable prediction model for missed HIV healthcare provider visits in a large United States clinical cohort. *Open Forum Infectious Diseases* 2021; 8: ofab130.
 203. Rebeiro PF, Jenkins CA, Bian A, Lake JE, Bourgi K, Moore RD, Horberg MA, Matthews WC, Silverberg MJ, Thorne J, Mayor AM, Lima VD, Palella FJ, Saag MS, Althoff KN, Gill MJ, Wong C, Klein MB, Crane HM, Marconi, VC, Shepherd BE, Sterling TR, Koethe JR. Risk of Incident Diabetes Mellitus, Weight Gain, and their Relationships with Integrase Inhibitor-based Initial Antiretroviral Therapy

- Among Persons with HIV in the US and Canada. *Clinical Infectious Diseases* 2021; 73: e2234–e2242.
204. Wudil UJ, Aliyu MH, Prigmore HL, Ingles DJ, Ahonkhai AA, Musa, BM, Muhammad H, Sani MU, Nalado AM, Abdu A, Abdussalam K, Shepherd BE, Dankishiya FS, Burgner AM, Ikizler TA, Wyatt CA, Kopp JB, Kimmel PL, Winkler CA, Wester CW. Apolipoprotein-L1 (APOL1) risk variants and associated kidney phenotypes in an adult HIV cohort in Nigeria. *Kidney International* 2021; 100: 146–154.
 205. ⁺Amorim G, Tao R, Lotspeich S, Shaw PA, Lumley T, Shepherd BE. Two-phase sampling designs for data validation in settings with covariate measurement error and continuous outcome. *Journal of the Royal Statistical Society, Series A* 2021; 184: 1368–1389.
 206. Sudenga SL, Lotspeich SC, Nyitray AG, Sirak B, Shepherd BE, Messina J, Sereday KA, Silva RC, Abrahamsen M, Baggio ML, Quiterio M, Lazcano-Ponce E, Villa L, Giuliano AR. The role of external genital lesions in HIV seroconversion among men participating in a multinational study. *Sexually Transmitted Diseases* 2022; 49: 55–58.
 207. Caro-Vega Y, Rebeiro PF, Shepherd BE, Belaunzaran-Zamudio PF, Crabtree-Ramirez B, Cesar C, Luz PM, Cortes CP, Padgett D, Gotuzzo E, McGowan CC, Sierra-Madero JG. Clinical effects of durability of immunosuppression in virologically suppressed ART-initiating persons with HIV in Latin America. A retrospective cohort study. *The Lancet Regional Health - Americas*. 2022; 8:100175.
 208. Coelho LE, Jenkins CA, Shepherd BE, Pape JW, Cordero FM, Padgett D, Crabtree-Ramirez B, Grinsztejn B, Althoff KN, Koethe JR, Marconi V, Tien P, Willig A, Moore RD, Castilho J, Colasanti J, Crane HM, Gill MJ, Horberg MA, Mayor A, Silverberg MJ, McGowan CC, Rebeiro PF. Weight gain post-ART in HIV+ Latinos/as differs in the US, Haiti, and Latin America. *The Lancet Regional Health - Americas* 2022; 8: 100173.
 209. Castilho JL, Bian A, Jenkins CA, Shepherd BE, Sigel K, Gill MJ, Kitahata MM, Silverberg MJ, Mayor AM, Coburn SB, Wiley D, Achenbach CJ, Marconi VC, Bosch RJ, Horberg HA, Rabkin CS, Napravnik S, Novak RM, Mathews WC, Thorne JE, Sun J, Althoff KN, Moore RD, Sterling TR, Sudenga SL. CD4/CD8 ratio and cancer risk among adults with HIV. *Journal of the National Cancer Institute* 2022; 114: 854–862.
 210. Schildcrout JS, Harrell FE, Heagerty PJ, Haneuse S, Di Gravio C, Garbett S, Rathouz PJ, Shepherd BE. Model-assisted analyses of longitudinal, ordinal outcomes with absorbing states. *Statistics in Medicine* 2022; 41: 2497–2512.
 211. Bhatta M, Bian A, Norwood J, Shepherd BE, Ransby I, Nelson J, Turner M, Sterling TR, Castilho JL. Low rates of contraception use in women with HIV. *Open Forum Infectious Diseases* 2022; 9: ofac113.

212. Wanjalla CN, Temu TM, Mashayekhi M, Warren CM, Shepherd BE, Gangula R, Fuseini H, Bailin S, Gabriel CL, Gangula P, Madhur MS, Kalams S, Mallal SA, Harrison DG, Beckman JA, Koethe JR. Interleukin-17A is associated with flow-mediated dilation and interleukin-4 with carotid plaque in persons with HIV. *AIDS* 2022; 36: 963–973.
213. Crabtree-Ramirez B, Jenkins CA, Shepherd BE, Jayathilake K, Veloso VG, Carrquiry G, Gotuzzo E, Cortes CP, Padgett D, McGowan CC, Sierra-Madero J, Koenig S, Pape JW, Sterling TR. Tuberculosis treatment intermittency in the continuation phase and mortality in HIV-positive persons receiving antiretroviral therapy. *BMC Infectious Diseases* 2022; 22: 341.
214. Ahonkhai AA, Rebeiro PF, Jenkins CA, Rickles M, Cook M, Conserve D, Pierce LJ, Shepherd BE, Brantley M, Wester C. Individual, Community, and Structural Factors Associated with Linkage to HIV Care Among People Diagnosed with HIV in Tennessee. *PLoS ONE* 2022; 17: e0264508.
215. *Eden S, Li C, Shepherd BE. Non-parametric estimation of Spearman’s rank correlation with bivariate survival data. *Biometrics* 2022; 78: 421–434.
216. Audet CM, Pettapiece-Phillips M, Tian Y, Shepherd BE, Vermund SH, Salato J. “If it weren’t for my traditional healer, I would be dead”: Engaging people living with HIV in rural Mozambique. *PLoS ONE* 2022; 17: e0270565.
217. Nunez I, Crabtree-Ramirez B, Shepherd BE, Sterling TR, Cahn P, Veloso VG, Cortes CP, Padgett D, Gotuzzo E, Sierra-Madero J, McGowan CC, Person AK, Caro-Vega Y. Late-onset opportunistic infections while receiving anti-retroviral therapy in Latin America: burden and risk factors. *International Journal of Infectious Diseases* 2022; 122: 469–475.
218. Norwood J, Kheshti AN, Shepherd BE, Rebeiro PF, Ahonkhai AA, Kelly SG, Wanjalla C. The impact of COVID-19 on the HIV care continuum in a large urban southern clinic. *AIDS and Behavior* 2022; 26: 2825–2829.
219. Wester CW, Shepherd BE, Wudil UJ, Musa BM, Ingles DJ, Prigmore HL, Dankishaya FS, Ahonkhai AA, Grema BA, Budge PJ, Takakura A, Olabisi OA, Winkler CA, Kopp JB, Bonventre JV, Wyatt CM, Aliyu MH. Etiology of persistent microalbuminuria in Nigeria: protocol and study design. *BMC Infectious Diseases* 2022; 22: 591.
220. Pierce LJ, Rebeiro P, Brantley M, Fields EL, Jenkins CA, Griffith D, Conserve D, Shepherd BE, Wester C, Ahonkhai AA. Who is not linking to HIV care in Tennessee – the benefits of an intersectional approach. *Journal of Racial and Ethnic Health Disparities* 2022; 9: 849–855.
221. Stalter RM, Amorim G, Mocello AR, Jakait B, Shepherd BE, Musick B, Bernard C, Bukusi EA, Wools-Kaloustian K, Cohen CR, Yiannoutsos CT, Patel RC. Contraceptive implant use duration is not associated with breakthrough pregnancy among

- women living with HIV and using efavirenz: a retrospective, longitudinal analysis. *Journal of the International AIDS Society* 2022; 25: e26001.
222. Anderson K, Pietersen E, Shepherd BE, Bian A, te Riele J, Cox H, Dheda K, Warren R, Sterling TR, van der Heijden YF. High mortality among patients hospitalized for drug-resistant tuberculosis with acquired second-line drug resistance and high HIV prevalence. *HIV Medicine* 2022; 23: 1085–1097.
 223. Desai N, Jenkins CA, Zanoni B, Nmoh A, Patel N, Shepherd BE, Hussen S, Doraivelu K, Pierce L, Carlucci J, Ahonkhah AA. High rates of viral suppression and care retention among youth born outside of the United States with perinatally-acquired HIV. *The Pediatric Infectious Diseases Journal* 2022; 41: 970–975.
 224. Werede AT, Terry JG, Nair S, Temu T, Shepherd BE, Bailin SS, Mashayekhi M, Gabriel CL, Lima M, Woodward B, Hanna L, Mallal SA, Beckman JA, Li J, Fajnzylber J, Harrison D, Carr JJ, Koethe JR, Wanjalla CN. Mean coronary cross-sectional area as a measure of arterial remodeling in persons with HIV. *Journal of the American Heart Association* 2022; 11: e025768.
 225. *Lotspeich SC, Shepherd BE, Amorim G, Shaw PA, Tao R. Efficient odds ratio estimation under two-phase sampling using error-prone data from a multi-national HIV research cohort. *Biometrics* 2022; 78: 1674–1685.
 226. Pietersen E, Anderson K, Cox H, Dheda K, Bian A, Shepherd BE, Sterling TR, Warren RM, van der Heijden YF. Variation in missed doses and reasons for discontinuation of anti-tuberculosis drugs during hospital treatment for drug-resistant tuberculosis in South Africa. *PLOS ONE* 2023; 18: e0281097.
 227. *Sack DE, Shepherd BE, Audet CM, de Schacht C, Samuels LR. Inverse probability weights for quasi-continuous ordinal exposures with a binary outcome: method comparison and case study. *American Journal of Epidemiology* 2023; 192: 1192–1206.
 228. Crockett KB, Schember CO, Bian A, Rebeiro PF, Keruly J, Mayer K, Mathews C, Moore RD, Crane H, Geng E, Napravnik S, Shepherd BE, Mugavero MJ, Turan B, Pettit AC. Relationships between Patient Race and Residential Race Context with Missed HIV Care Visits in the United States, 2010-2015. *Clinical Infectious Diseases* 2023; 76: 2163–2170.
 229. Person AK, Crabtree-Ramirez B, Kim A, Veloso V, Maruri F, Wandeler G, Fox M, Moore RD, Gill MJ, Imran D, Van Nguyen K, Nalitya E, Muyindike W, Shepherd BE, McGowan CC. Cryptococcal Meningitis and Clinical Outcomes in Persons with HIV: A Global View. *Clinical Infectious Diseases* 2023; 76: 2116–2125.
 230. †Shepherd BE, Han K, Chen T, Bian A, Pugh SK, Duda SN, Lumley T, Heerman WJ, Shaw PA. Multi-wave validation sampling for error-prone electronic health records. *Biometrics* 2023; 79: 2649–2663.

231. Sack DE, Matino A, Graves E, Emilio A, Shepherd BE, De Schacht C, Audet CM. Correlated depressive symptoms within seroconcordant, expectant partners living with HIV in Zambezia Province, Mozambique: a cross-sectional study. *AIDS Care* 2023; 35: 1732–1740.
232. *+Tu S, Li C, Zeng D, Shepherd BE. Rank intraclass correlation for clustered data. *Statistics in Medicine* 2023; 42: 4333–4348.
233. Shepherd BE, Hussaini N, Ingles D, Cassell H, Wester CW, Aliyu M. Training the next generation of biostatisticians in West Africa: the Vanderbilt Nigeria Biostatistics Training Program (VN-BioStat). *Journal of Global Health Reports* 2023; 7: e2023067.
234. Lotspeich SC, Shepherd BE, Kariuki MA, Wools Kaloustian K, McGowan CC, Musick B, Semeere A, Crabtree-Ramirez BE, Mkwashapi DM, Cesar C, Ssemakadde M, Machado DM, Ngeresa A, Ferreira FGF, Lwali J, Marcelin A, Cardoso SW, Luque MT, Otero L, Cortes CP, Duda SN. Lessons learned from over a decade of data audits in international observational HIV cohorts in Latin America and East Africa. *Journal of Clinical and Translational Science* 2023;7:e245.
235. Li C, Tian Y, Zeng D, Shepherd BE. Asymptotic properties for cumulative probability models for continuous outcomes. *Mathematics* 2023; 11: 4896.
236. *Eden S, Li C, Shepherd BE. Spearman-like correlation measure adjusting for covariates in bivariate survival data. *Biometrical Journal* 2023; 65: 2200137.
237. *Tian Y, Shepherd BE, Li C, Zeng D, Schildcrout JS. Analyzing clustered continuous response variables with ordinal regression models. *Biometrics* 2023; 79: 3764–3777.
238. Shepherd BE, Hussaini N, Huang A, Van Wyk C, Kowalski MS, Ingles DJ, Wester CW, Li C, Aliyu MH. The Vanderbilt Nigeria Biostatistics Training Program (VN-BioStat): results from a skills workshop. *International Journal of Statistics and Probability* 2023; 12: 66–72.
239. Amorim G, Tao R, Lotspeich SC, Shaw PA, Lumley T, Patel RC, Shepherd BE. Three-phase generalized raking and multiple imputation estimators to address error-prone data. *Statistics in Medicine* 2024; 43: 379–394.
240. Brown LL, Perkins J, Shepherd BE, Ramasamy S, Wilkins M, Osman A, Turner M, Link T, Edgerton R, Suiter SV, Pettit AC. Piloting safety and stabilization: a multi-component trauma intervention to improve HIV viral suppression, retention in care, and post-traumatic stress disorder in a southern United States HIV service organization. *AIDS in Behavior* 2024; 28: 174–185.
241. Ahonkhai AA, Bian A, Robbins NN, Maurer LA, Clouse K, Pierce LJ, Perkins JM, Wernke SA, Shepherd BE, Brantley M. Characterizing residential mobility among people living with HIV in Tennessee and its impact on HIV care outcomes. *AIDS* 2024; 38: 397–405.

242. Castillo-Rozas G, Tu S, Mejia F, Rouzier V, Luz PM, Shepherd BE, Cortes CP. Clinical outcomes and risk factors for immune recovery and all-cause mortality in Latin Americans living with HIV with virological success: a retrospective cohort study. *Journal of the International AIDS Society* 2024; 27: e26214.
243. Sneed NM, Heerman WJ, Shaw PA, Han K, Chen T, Bian A, Pugh SK, Duda SN, Lumley T, Shepherd BE. Associations between gestational weight gain, gestational diabetes, and childhood obesity incidence. *Maternal and Child Health Journal* 2024; 28: 372–381.
244. *#Lotspeich SC, Amorim G, Shaw PA, Tao R, Shepherd BE. Optimal multi-wave validation of secondary use data with outcome and exposure misclassification. *The Canadian Journal of Statistics* 2024; 52: 532–554.
245. Audet CM, Graves E, Shepherd BE, Prigmore HL, Brooks HL, Emilio A, Matino A, Paulo P, Diemer MA, Frisby M, Sack DE, Aboobacar A, Barreto E, Van Rompaey S, De Shacht C. Partner-based HIV treatment for seroconcordant couples attending antenatal and postnatal care in rural Mozambique: a cluster randomized controlled trial. *Journal of Acquired Immune Deficiency Syndromes* 2024; 96: 259–269.
246. *Tian Y, Li C, Tu S, James NT, Harrell FE, Shepherd BE. Addressing multiple detection limits with semiparametric cumulative probability models. *Journal of the American Statistical Association* 2024; 119: 864–874.
247. Norwood J, Jenkins CA, Bhatta M, Turner M, Bian A, Nelson J, Ransby I, Hughes D, Koethe JR, Sterling TR, Shepherd BE, Castilho JL. Hormonal contraception and risk of cardiometabolic disease in women with HIV. *Journal of Women's Health* 2024; 33: 966–974.
248. Alexander L, Brown V, Koethe JR, Jenkins CA, Shepherd BE, Green M, Myers HF, Chandler CJ. Keeping it real: an exploratory analysis of HIV testing history, PrEP awareness, and PrEP use among black cisgender women in Middle Tennessee, 2018. *Journal of the National Medical Association* 2024; 116: 338–350.
249. Rebeiro PF, Thome JC, Gange SJ, Althoff KN, Berry SA, Horberg MA, Moore RD, Silverberg MJ, Sack DE, Sterling TR, Sant'Anna P, Shepherd BE. The impact of Medicaid expansion under the Affordable Care Act on HIV care continuum outcomes across the United States. *Health Affairs Scholar* 2024; 2: qxae128.
250. Kana SA, Alkassim RS, Adamu AL, Ibrahim IY, Shepherd BE, Sani MU, Wester CW, Aliyu MH. Prevalence and predictors of right ventricular dysfunction among adults living with HIV in northwest Nigeria. *Cardiovascular Diagnosis and Therapy* 2024; 14: 753–770.
251. Castilho JL, Fonseca FF, Kim A, Jalil E, Tu S, Beber AMB, Benzaken AS, Veloso VG, Grinsztejn B, Shepherd BE, Miranda AEB. Prenatal syphilis and adverse pregnancy outcomes in women with HIV receiving ART in Brazil. *The Lancet Regional Health Americas* 2024; 39: 100894.

252. *Tu S, Li C, Shepherd BE. Between- and within-cluster Spearman rank correlation. *Statistics in Medicine* 2025; 44: e10326.
253. Castillo-Rozas G, Fonseca FF, Castilho JL, Rebeiro PF, Machado DM, Luque MT, Jalil EM, Mejia F, Kim A, Shepherd BE, Cortes CP. Antiretroviral therapy to prevent vertical transmission in Latin America: Where are we in the Treat-All era? *Journal of the International AIDS Society* 2025; 28: e26411.

In Press

254. *Abdulhamid A, Shepherd BE, Wudil U, Van Wyk C, Dankishiya F, Hussaini N, Wester CW, Aliyu M. Sick cell trait, APOL1 risk allele status and chronic kidney disease among ART-experienced adults living with HIV in northern Nigeria. *International Journal of STD and AIDS* (in press).
255. Yang JB, Shepherd BE, Lumley T, Shaw PA. Optimum allocation for adaptive multi-wave sampling in R: the R package ‘optimall’. *Journal of Statistical Software* (in press).
256. Fonseca FF, Ranadive P, Shepherd BE, Ferreira FGF, Rodriguez MF, Machado DM, Rouzier V, Varela D, Maruri F, Rebeiro PF, Grinsztejn B, Cardoso SW, Veloso VG, Castilho JL, Jalil EM, CCASAnet. Persistent sex disparities in access to dolutegravir-based antiretroviral therapy in Latin America and the Caribbean: results from a retrospective observational study using data from 2017 to 2022. *Journal of the International AIDS Society* (in press).
257. Amorim G, Tao R, Lumley T, Shaw PA, Shepherd BE. Ascertainment conditional maximum likelihood for continuous outcome under two-phase response-selective design. *Statistics in Medicine* (in press).
258. Machado DM, Duda SN, Succi RCM, Kim A, Ranadive P, Rouzier V, Crabtree-Ramírez B, Luque MT, Mejia F, Rodriguez F, Pinto J, Cardoso SW, Maruri F, Shepherd BE, McGowan CC, Person AK. HIV care continuum outcomes among adolescents and young adults living with HIV in Latin America and the Caribbean: association with depression and substance use. *Open Forum Infectious Diseases* (in press).

* First author is student supervised by Dr. Shepherd.

+ Won the *Methods Publication Award* for the Vanderbilt Department of Biostatistics, given annually to recognize one paper published by department members. Recipient in 2016, 2017, 2021, and 2023.

** Won the 2016 *Canadian Journal of Statistics Award*, given annually by the Statistical Society of Canada to one paper published in *CJS*.

Won the David P. Byar Early Career Award for 2023, given to one paper led by an early career investigator (Sarah Lotspeich) in the Biometrics section of the American Statistical Association.

† Won Best Paper in *Biometrics* by an IBS Member in 2023.

Peer-Reviewed Report

Shepherd BE, Shaw PA. New methods to improve data accuracy in studies using electronic health record data. Washington (DC): Patient-Centered Outcomes Research Institute (PCORI); 2022 August. PCORI Final Research Reports.
<https://doi.org/10.25302/08.2022.me.160936207>

Invited Discussion

Shepherd BE, Liu Q. Discussion of ‘Regularized regression for categorical data.’ *Statistical Modelling* 2016; 16: 238–248.

Liu D, Shepherd BE, Discussion on ‘Assessing the goodness of fit of logistic regression models in large samples: a modification of the Hosmer-Lemeshow test’ by Giovanni Nattino, Michael L. Pennell, and Stanley Lemeshow. *Biometrics* 2020; 76: 572–574.

Book Chapter

Shepherd BE, Liu Q, Wanga V, Li C. Probability-scale residuals in HIV/AIDS research: diagnostics and inference. *Quantitative Methods for HIV/AIDS Research*, editors Chan C, Hudgens MG, Chow SC. 2017. Chapman and Hall/CRC.

Book Reviews

Shepherd BE. Review of “Global Sensitivity Analysis. The Primer” by Saltelli A et al. *Biometrics* 2009; 65: 1311–1312.

Shepherd BE, Jarrett R, Fu L. Review of “Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction” by Imbens G and Rubin D. *Biometrics* 2016; 72: 1387–1388.

Letter to the Editor

Morel KR, Shepherd BE (2008). Meta-analysis of the Morel Emotional Numbing Test for PTSD: comment on Sing, Avasthi, and Grover. *German Journal of Psychiatry* 11: 128–131.

Presentations

Invited

1. An analysis problem arising in HIV vaccine trials and our proposed solution. Department of Statistics, Brigham Young University, February 28, 2006.

2. Comparing outcomes that only exist in a group chosen after randomization.
Causal Inference Working Group, Department of Biostatistics, Johns Hopkins University, December 14, 2006.
3. Design of medical studies and some analysis advice.
The Caribbean, Central, and South American network, Educational Visit, Huesped Foundation, Buenos Aires, Argentina, September 26, 2007.
4. Kaplan-Meier survival estimates.
The Caribbean, Central, and South American network, Educational Visit, Hospital Universitario Clementino Fraga Filho, Rio de Janeiro, Brazil, November 7, 2007.
5. Assessing causal effects in the doomed.
Western North American Region of the International Biometric Society, University of California, Davis, June 25, 2008.
6. Vaccine efficacy among the doomed when monotonicity is unreasonable.
Joint Statistical Meetings. Denver. August 5, 2008.
7. Assessing the effect of vaccination in the doomed.
Department of Health Studies, University of Chicago, October 22, 2008.
8. The cost of checking proportional hazards.
Biostatistics Research Branch, National Institute of Allergy and Infectious Diseases, National Institutes of Health, November 14, 2008.
9. Data analysis: science and art.
Workshop on behavior intervention for people with high-risk for AIDS, National Center for STD/AIDS Control and Prevention, China Centers for Disease Control, Beijing, China, April 25, 2009.
10. Assessing the effect of vaccination on post-infection outcomes.
Department of Economics, Indiana University Purdue University Indianapolis, October 15, 2009.
11. Test of association between two ordinal variables while adjusting for covariates.
Department of Biostatistics, Johns Hopkins University, October 21, 2009.
12. The Caribbean, Central and South America Network for HIV Epidemiology: outcomes and errors.
Infectious Diseases / Center for AIDS Research Conference, University of North Carolina, Chapel Hill, November 6, 2009.
13. Estimating the optimal CD4 count for HIV-infected individuals to start antiretroviral therapy.
Causal Inference Group, Department of Epidemiology, University of North Carolina, Chapel Hill, November 6, 2009.
14. Eliciting a counterfactual sensitivity parameter.
Atlantic Causal Inference Conference, New York City, May 20, 2010.

15. Sensitivity analyses to assess the effect of vaccination on those who would have been infected regardless of treatment assignment.
Western North American Region of the International Biometric Society, University of Washington, June 21, 2010.
16. Estimating the optimal CD4 count for HIV-infected individuals to start antiretroviral therapy.
Division of Epidemiology, University of Utah, July 6, 2010.
17. A new residual for ordinal outcomes and a new test of association for ordinal predictors.
Department of Biostatistics and Epidemiology, University of Pennsylvania, April 12, 2011.
18. Accounting in analysis for data errors discovered through sampling.
University of Pennsylvania Annual Conference on Statistical Issues in Clinical Trials, Philadelphia, April 13, 2011.
19. Sensitivity analysis of per-protocol treatment efficacy in randomized blinded trials.
Workshop on Causal Inference in Health Research, Centre de Recherches Mathématiques, Université de Montréal, May 11, 2011.
20. Practical applications of principal stratification in HIV research.
Eastern North American Region of the International Biometric Society, Washington DC, April 3, 2012.
21. Discussion of four papers on ‘Mismeasurement of time-to-event outcomes: issues and methods of analysis.’
Joint Statistical Meetings, Montreal, August 4, 2013.
22. Comparing results from two methods for estimating the optimal CD4 threshold for initiating antiretroviral therapy.
Joint Statistical Meetings, Montreal, August 8, 2013.
23. Comparing different methods for assessing when to start treatment.
Division of Biostatistics, University of Minnesota, December 4, 2013.
24. Methods to use audit data in observational studies.
Joint Statistical Meetings, Boston, August 5, 2014.
25. Covariate-adjusted Spearman’s rank correlation with probability-scale residuals.
School of Mathematics and Statistics, Victoria University of Wellington, New Zealand, February 2, 2016.
26. Covariate-adjusted Spearman’s rank correlation with probability-scale residuals.
Department of Epidemiology and Biostatistics, Case Western Reserve University, February 17, 2016.

27. Covariate-adjusted Spearman's rank correlation with probability-scale residuals.
Department of Bioinformatics and Biostatistics, University of Louisville, February 19, 2016.
28. Let's make America great again: principal stratification is the principal way forward.
(pro side of debate)
Epidemiology Congress of the Americas, Miami, June 24, 2016.
29. Covariate-adjusted Spearman's rank correlation with probability-scale residuals.
Biostat and Biomath Seminar, Fred Hutchinson Cancer Research Center, 5 July, 2016.
30. Covariate-adjusted Spearman's rank correlation with probability-scale residuals.
International Chinese Statistical Association Conference, Shanghai, China, December 20, 2016.
31. Probability-scale residuals for continuous, discrete, and censored data.
Canadian Journal of Statistics Award Address. 45th Annual Meeting of the Statistical Society of Canada, Winnipeg, Manitoba, June 14, 2017.
32. Lengthening TB therapy in persons co-infected with HIV and tuberculosis: Challenges and considerations for designing a multi-national pragmatic trial.
Workshop on Risk Modeling, Management and Mitigation in Health Sciences. Centre de Recherches Mathematiques. University of Montreal, December 12, 2017.
33. Correcting for errors in variables derived from electronic health records using validation sampling and multiple imputation.
Joint Statistical Meetings, Vancouver, July 31, 2018.
34. Opportunities and challenges using data from large observational HIV cohorts.
Workshop on Innovative Data Driven Strategies for Infectious Diseases. National Institute of Allergy and Infectious Diseases. Rockville, Maryland, September 24, 2018.
35. Covariate-adjusted Spearman's rank correlation with probability-scale residuals.
Emory University, November 8, 2018.
36. Correcting for errors in variables derived from electronic health records using validation sampling and multiple imputation.
UC Davis, February 5, 2019.
37. Errors in multiple variables in HIV cohort and EHR data: statistical challenges and opportunities.
Workshop on Statistical Challenges and Opportunities in HIV/AIDS Research in the Era of Getting-to-Zero HIV infections. National Institute of Allergy and Infectious Diseases. Philadelphia, March 23, 2019.
38. Multiple imputation to address errors in data errors in electronic health record analyses: advantages and disadvantages.
ENAR, Philadelphia, March 27, 2019.

39. Modeling continuous response variables using ordinal regression.
UC San Diego, June 5, 2019.
40. A pragmatic approach approach for reproducible research with sensitive data.
Society for Epidemiologic Research, Minneapolis, Minnesota, June 20, 2019.
41. Errors in electronic health records: what two-phase sampling teaches us about data validation.
Joint Statistical Meetings, Denver, Colorado, July 30, 2019.
42. Covariate-adjusted Spearman's rank correlation with probability-scale residuals.
Washington University, September 20, 2019.
43. Addressing outcome detection limits using semiparametric cumulative probability models.
ENAR, Nashville (virtual), March 24, 2020.
44. Analyzing continuous longitudinal response data with ordinal regression models.
Joint Statistical Meetings, Philadelphia (virtual), August 6, 2020.
45. Statistical approaches to improve HIV studies with error-prone data.
Emory Center for AIDS Research, Atlanta (virtual), February 18, 2021.
46. Statistical approaches to improve studies with error-prone data.
The Ohio State University, Columbus (virtual), March 12, 2021.
47. Analyzing continuous response data with ordinal cumulative probability models.
Harvard, Boston (virtual), November 5, 2021.
48. Fitting semiparametric linear transformation models with ordinal regression models.
ENAR, Houston, March 29, 2022.
49. Strengthening biostatistics resources at Aminu Kano Teaching Hospital and Bayero University Kano, Nigeria in collaboration with Vanderbilt University, USA.
Strengthening Biostatistics Resources in sub-Saharan Africa: Networking and Collaborations. Banjul, Gambia (virtual), September 1, 2022.
50. Multi-wave validation sampling to improve estimates derived from electronic health record data.
ENAR, Nashville, March 20, 2022.
51. The Vanderbilt-Nigeria Biostatistics Training Program.
Center for AIDS Research Symposium on Statistics and Data Science in HIV. Brown University, Providence, RI. June 6, 2023.
52. Statistical methods and designs for addressing data quality in studies of HIV/AIDS.
Division of Biostatistics, University of Utah, June 26, 2023.

53. Multi-wave validation sampling to improve estimates derived from electronic health record data.
FDA Sentinel Innovation Center (virtual), July 11, 2023.
54. Addressing multiple detection limits with semiparametric cumulative probability models.
School of Statistics, University of Minnesota, September 27, 2023.
55. Statistical methods and designs to address data quality in studies that use electronic health record data.
Department of Biostatistics and Epidemiology, Memorial Sloan Kettering Cancer Center, October 4, 2023.
56. Synthetic HIV cohort data to bridge open science and privacy protections.
International Chinese Statistical Association Symposium, Nashville, TN, June 17, 2024.
57. Multi-wave validation sampling for error-prone electronic health records.
Showcase Session, International Biometric Conference, Atlanta, GA, December 12, 2024.
58. Design and analysis of a multi-wave two phase study to address errors in a multinational HIV research network.
International Joint Conference CFE-CMStatistics, London, UK (virtual), December 16, 2024.
59. Design and analysis of a multi-wave two-phase study to address data errors in a multinational HIV research network.
ENAR, New Orleans, March 24, 2025.
60. Synthetic data.
International Workshop on HIV and Hepatitis Observational Databases, Toledo, Spain, March 29, 2025.
61. Synthetic data.
African Population Health Research Center, Nairobi, Kenya (virtual), April 24, 2025.

Contributed

1. Shepherd BE, O'Neill K. Analysis of tail moments in the comet assay. American Microbiology Association. Idaho Falls. May 28, 1998.
2. Shepherd BE, Schaalje GB. Statistical analysis of the comet assay using a mixture of gamma distributions. 42nd Annual Conference on Cardiovascular Disease Epidemiology and Prevention. Honolulu. April 23, 2002.

3. Shepherd BE, Gilbert PB. Causal inference in HIV vaccine trials: Comparing biomarkers measured only in a subgroup chosen post-randomization. Joint Statistical Meetings. San Francisco. August 5, 2003.
4. Shepherd BE, Guttorp P, Abkowitz JL. Estimating human hematopoietic stem cell kinetics using granulocyte telomere lengths. American Society of Hematology. San Diego. December 7, 2003.
5. Shepherd BE, Gilbert PB, Gurwith M, Francis D. Assessment of HIV vaccine effects on viral load in the first Phase III trial of AIDSVAX B/B. Eleventh Conference on Retroviruses and Opportunistic Infections. San Francisco. February 11, 2004.
6. Shepherd BE. Comparing outcomes that only exist in a group chosen after randomization. Department of Biostatistics, UCSF, April 6, 2005.
7. Shepherd BE. Comparing outcomes that only exist in a group chosen after randomization. Department of Statistics, Simon Fraser University, March 7, 2005.
8. Shepherd BE. Comparing outcomes that only exist in a group chosen after randomization. Department of Biostatistics, Harvard University, March 3, 2005.
9. Shepherd BE. Comparing outcomes that only exist in a group chosen after randomization. Statistics Group, RAND Corporation, February 21, 2005.
10. Shepherd BE. Comparing outcomes that only exist in a group chosen after randomization. Department of Biostatistics, Emory University, February 18, 2005.
11. Shepherd BE. Comparing outcomes that only exist in a group chosen after randomization. Department of Biostatistics, Vanderbilt University, January 19, 2005.
12. Shepherd BE, Kiem HP, Dunbar CE, Larochelle A, Donahue RE, Seggewiss R, Lansdorp PM, Guttorp P, Abkowitz JL. Estimating the replication rate of hematopoietic stem cells in non-human primates: a test of Hayflick's hypothesis. American Society of Hematology. Atlanta. December 10, 2005.
13. Shepherd BE. The causal effect of finasteride on the severity of prostate cancer. Joint Statistical Meetings. Seattle. August 7, 2006.
14. Shepherd BE, Sterling TR, Moore RD, Raffanti SP, Hulgán T. Evaluation of a prognostic model for HIV disease progression after initiation of HAART. 11th International Workshop on HIV Observational Databases. Monte Carlo, Monaco. March 23, 2007.
15. Shepherd BE. Modelling HIV disease progression among persons initiating HAART: the cost of checking and correcting violations of the proportional hazards assumption. 11th International Workshop on HIV Observational Databases. Monte Carlo, Monaco. March 23, 2007.
16. Shepherd BE. The cost of checking proportional hazards. Joint Statistical Meetings. Salt Lake City. August 1, 2007.

17. Shepherd BE, Jenkins CA, Rebeiro PF, Stinnette SE, Bebawy SS, Kheshti A, Blackwell R, Raffanti S, Hulan T, Sterling TR. When to start highly active antiretroviral therapy: a pilot study of inverse probability weighted methods. 12th International Workshop on HIV Observational Databases. Malaga, Spain. March 29, 2008.
18. Shepherd BE, Jenkins CA, Rebeiro PF, Stinnette SE, Bebawy SS, McGowan CC, Hulan T, Sterling TR. When to start highly active antiretroviral therapy: estimates of an optimal rule based on CD4 count. 13th International Workshop on HIV Observational Databases. Lisbon, Portugal. March 26, 2009.
19. Shepherd BE, Yu C. Statistical methods to account for data errors discovered from an audit. Joint Statistical Meetings. Washington DC. August 4, 2009.
20. Shepherd BE, Jenkins CA, Rebeiro PF, Stinnette SE, McGowan CC, Hulan T, Sterling TR. Comparing three methods for estimating the optimal CD4-threshold for HAART initiation. 14th International Workshop on HIV Observational Databases. Sitges, Spain. March 26, 2010.
21. Shepherd BE, Li C. Tests for conditional association between an ordinal predictor and continuous, binomial, or ordinal outcomes. Joint Statistical Meetings. Vancouver, BC. August 1, 2010.
22. Shepherd BE. Statistical methods to account for data errors discovered from an audit. Executive Committee Meeting for the International Epidemiologic Databases to Evaluate AIDS. Boston, MA. March 3, 2011.
23. Shepherd BE, Li C. Methods for leveraging the order information in ordered categorical variables. 15th International Workshop on HIV Observational Databases. Prague, Czech Republic. March 25, 2011.
24. Shepherd BE, Jenkins CA, Sterling TR for the ART-CC. Higher rates of AIDS during the first year of therapy among those born outside Europe, the United States, and Canada. 16th International Workshop on HIV Observational Databases. Athens, Greece. March 29, 2012.
25. Shepherd BE. Multiple imputation methods to account for data errors discovered by an audit. 16th International Workshop on HIV Observational Databases. Athens, Greece. March 30, 2012.
26. Shepherd BE, Giganti MJ. Comparing different approaches to account for cohort heterogeneity in the Cox proportional hazards model. 17th International Workshop on HIV Observational Databases. Cavtat, Croatia. April 11, 2013.
27. Shepherd BE. Assessing and interpreting the association between continuous covariates and outcomes using splines. 19th International Workshop on HIV Observational Databases. Catania, Sicily. March 27, 2015.

28. Shepherd BE, Jenkins C, Luque M, Cesar C, Rouzier V, Machado D, Maia M, Padgett D, Pinto J, McGowan C. 19th International Workshop on HIV Observational Databases. Mortality and regimen change among children starting antiretrovirals in Latin America. Catania, Sicily. March 27, 2015.
29. Shepherd BE, Li C, Liu Q. Probability-scale residuals for continuous, discrete, and censored data. Joint Statistical Meetings. Seattle, WA. August 9, 2015.
30. Shepherd BE, Blevins M. Reproducible research with observational HIV cohort data. 20th International Workshop on HIV Observational Databases. Budapest, Hungary. April 8, 2016.
31. Shepherd BE, Liu Q, Li C, Harrell FE. Analyzing continuous data with ordinal regression models. International Workshop on HIV and Hepatitis Observational Databases. Fuengirola, Spain. March 23, 2018.
32. Amorim G, Tao R, Lotspeich S, Shaw P, Patel R, Shepherd BE. Three-phase generalized raking to address errors in routinely collected data. International Workshop on HIV and Hepatitis Observational Databases. Sevilla, Spain. March 24, 2022.
33. Shepherd BE. Design and analysis of a multi-wave two phase study that addresses data errors in a large observational HIV network. Joint Statistical Meetings. Toronto, CA. August 8, 2023.